Chapter 2

Diode and its applications

Diode

- ✓ Is a two terminal(anode & cathode) device.
- ✓ Is a non-linear device.
- ✓ Is a device that conducts current in only one direction.
- ✓ Is formed when a piece of semiconductor material (of the same base Si or Ge) half of which is doped by P-type impurity and the other half by N-type impurity.
- ✓ The plane dividing the two zones is called a junction (PN-junction).
- ✓ Theoretically, junction plane is assumed to lie where density of donors and acceptors is equal.

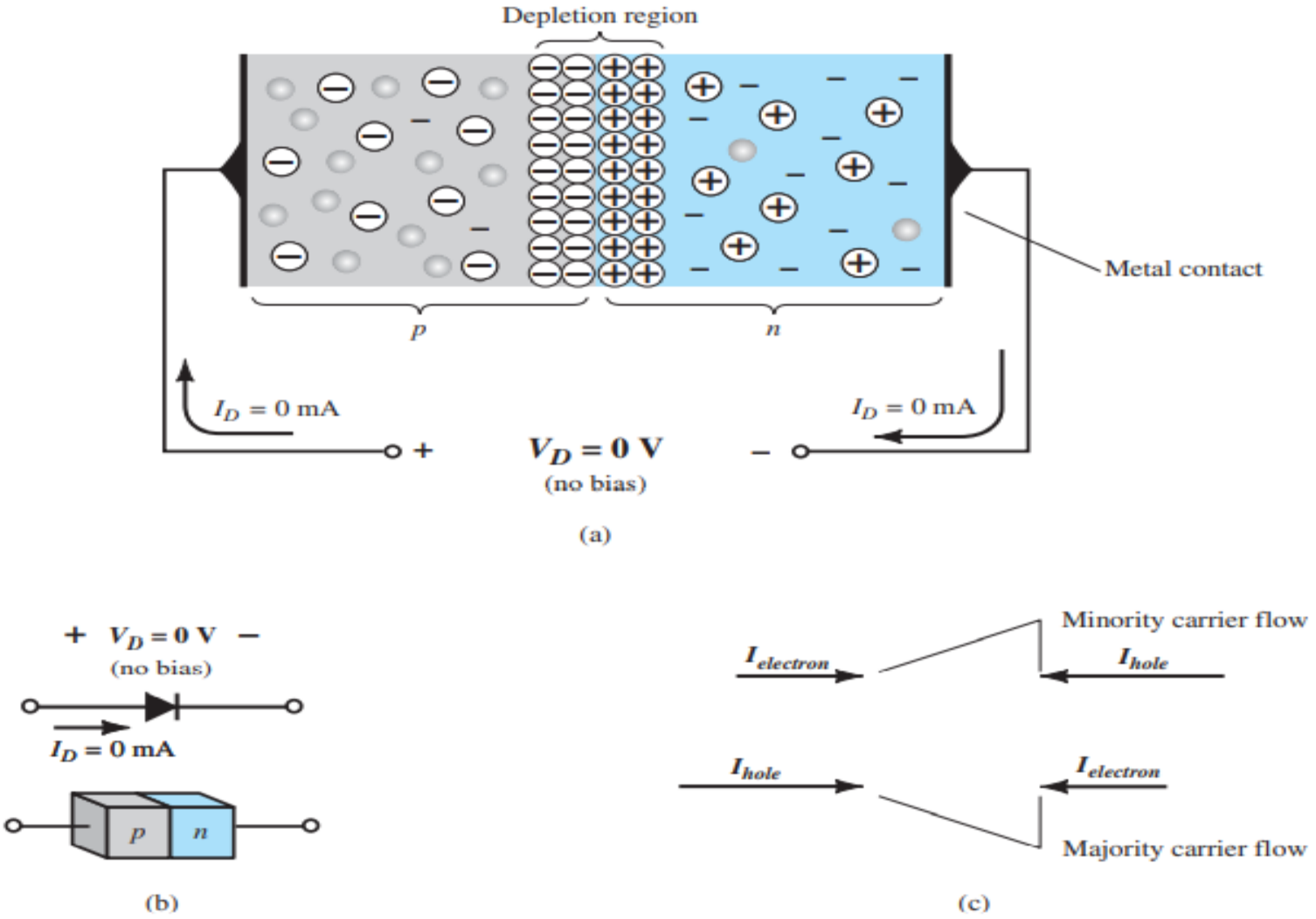
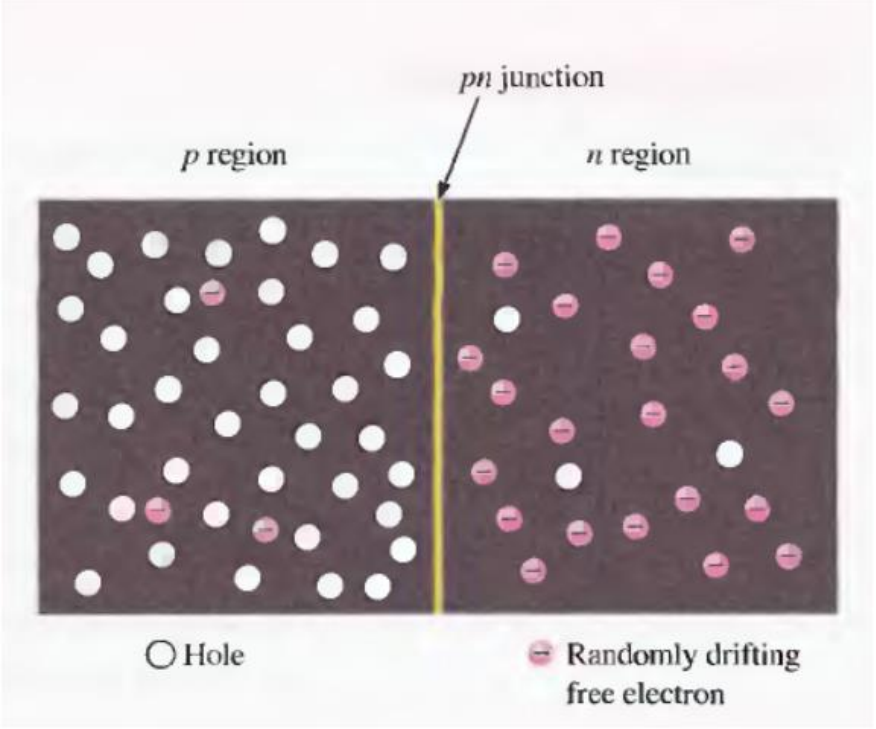


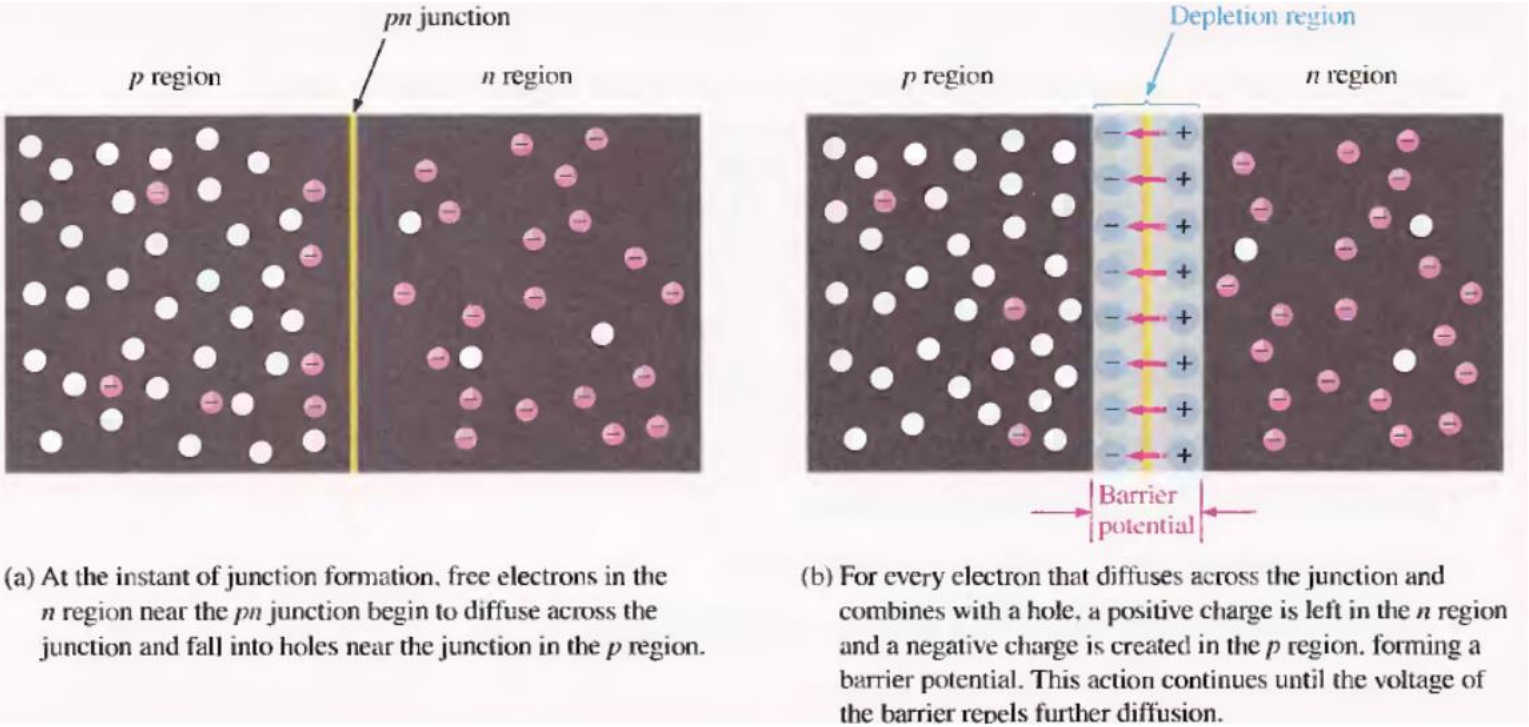
Fig.1 A p–n junction with no external bias: (a) an internal distribution of charge; (b) a diode symbol, with the defined polarity and the current direction; (c) demonstration that the net carrier flow is zero at the external terminal of the device when V D = 0 V.

Formation of the Depletion Region

- ✓ The free electrons in the n region are randomly drifting in all directions.
- ✓ At the instant of the pn junction formation, the free electrons near the junction in the n region begin to diffuse across the junction into the p region where they combine with holes near the junction.



Fig



(a) At the instant of junction formation, free electrons in the *n* region near the *pn* junction begin to diffuse across the junction and fall into holes near the junction in the *p* region.

(b) For every electron that diffuses across the junction and combines with a hole, a positive charge is left in the *n* region and a negative charge is created in the *p* region, forming a barrier potential. This action continues until the voltage of the barrier repels further diffusion.

By this process

1) n-region loses free electron as they diffuses across the junction and creates a layer of positive ion.
2) The p-region loses holes as the electrons and holes combine and creates a layer of negative ion near the junction.
3) As electrons continue to diffuse across the junction more and more positive and negative ions are created near the junction and depletion region is formed. At a point where this immobile ions refuse the diffusion of free carriers, an equilibrium is reached.

Biasing

Forward biased pn-junction

- ✓ A forward bias (ON) condition is established by applying (connecting) the +ve potential to the p-type material and –ve potential to the n-type material.

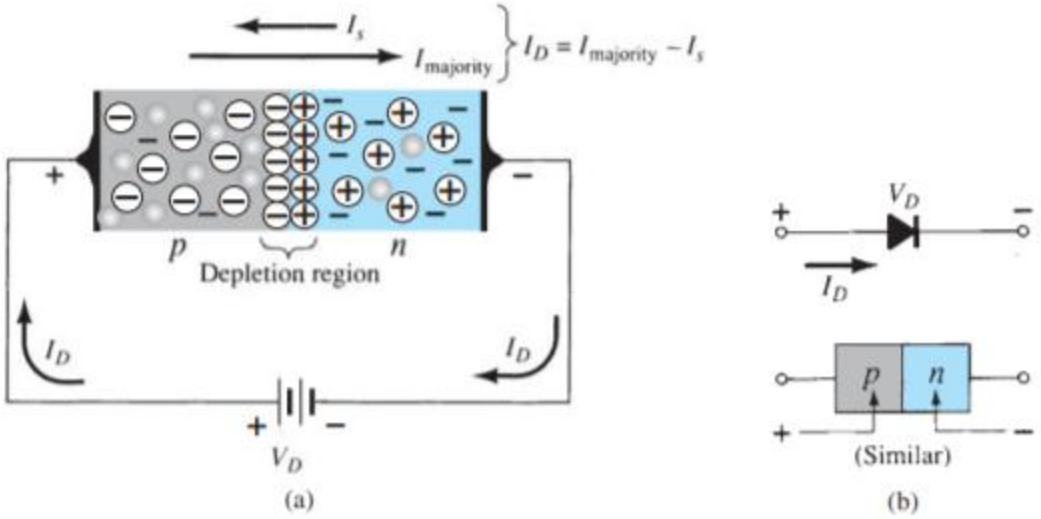


Fig. Forward-biased p–n junction: (a) internal distribution of charge under forward-bias conditions; (b) forward-bias polarity and direction of resulting current.

- ✓ The application of forward bias potential VD will pressure electrons in the N-type material and holes in the p-type material to recombine with the ion near the boundary and reduce the width of depletion region.
- ✓ The junction offers low resistance in the forward direction.

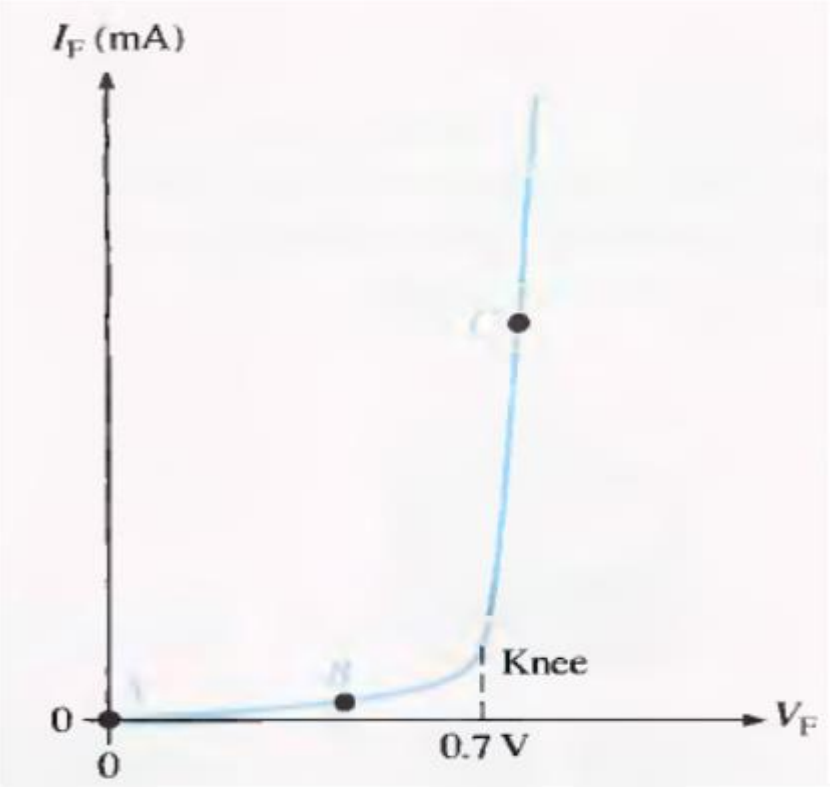


Fig.VI-characteristic of diode(forward biased)

Reverse bias condition (VD<0v)-off state

- ✓ If an external potential of v volts is applied across the pn-junction such that the positive terminal is connected to the n-type material and the –ve terminal is connected to the p-type material reverse biasing occur.

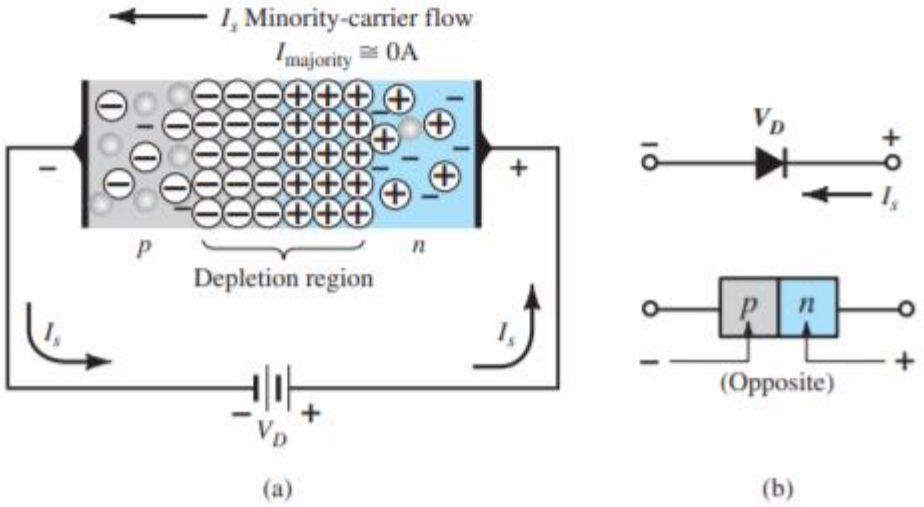


Fig Reverse-biased p–n junction: (a) internal distribution of charge under reverse-bias conditions; (b) reverse-bias polarity and direction of reverse saturation current.

- ✓ In this case, holes are attracted by negative battery terminal and electrons by the positive terminal; so that, both holes and electrons move away from the junction and away from each other.
- ✓ The number of uncovered ions (immobile ions) in the depletion region increases.
- ✓ The junction offers high resistance because of increase in barrier potential.
- ✓ When the diode is reversely biased, majority carriers are blocked and only a small current (due to minority carriers) flows through the diode.
- ✓ As the reverse voltage increased from zero, the reverse current very quickly reaches its maximum or saturation value (Is) which is known as leakage current.
- ✓ The value of (Is) is independent of the applied reverse voltage but depend on:-
  i. Temperature
  ii. Degree of doping
  iii. Physical size of the junction…etc

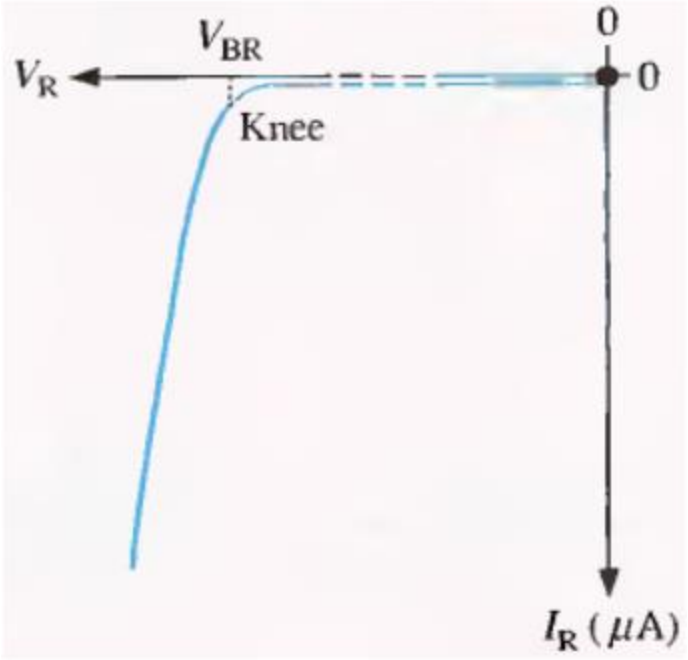


Fig.VI-characteristic of diode(reverse biased)

Reverse breakdown

Zener breakdown (Higher electric field)

- ✓ Zener breakdown occurs in junction which, being heavily doped, and have narrow depletion layers. (tunneling of electrons from VB to CB).
- ✓ Breakdown voltage =>sets up strong electric field across this narrow layer.
- ✓ Strong electric field =>breaks covalent bonds =>EHP is created. ↑reverse voltage=produces large number of current carriers.
- ✓ Therefore, the junction has low resistance in breakdown region.

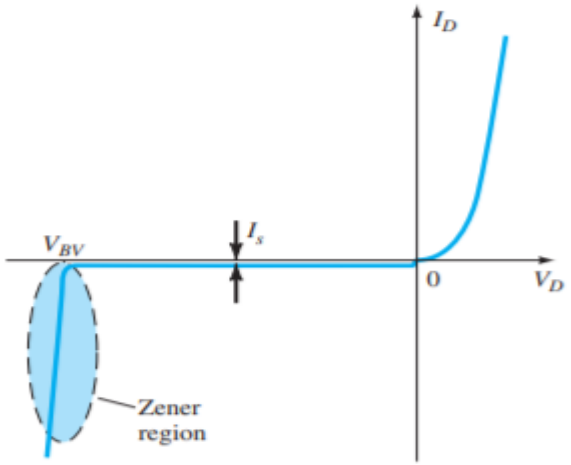
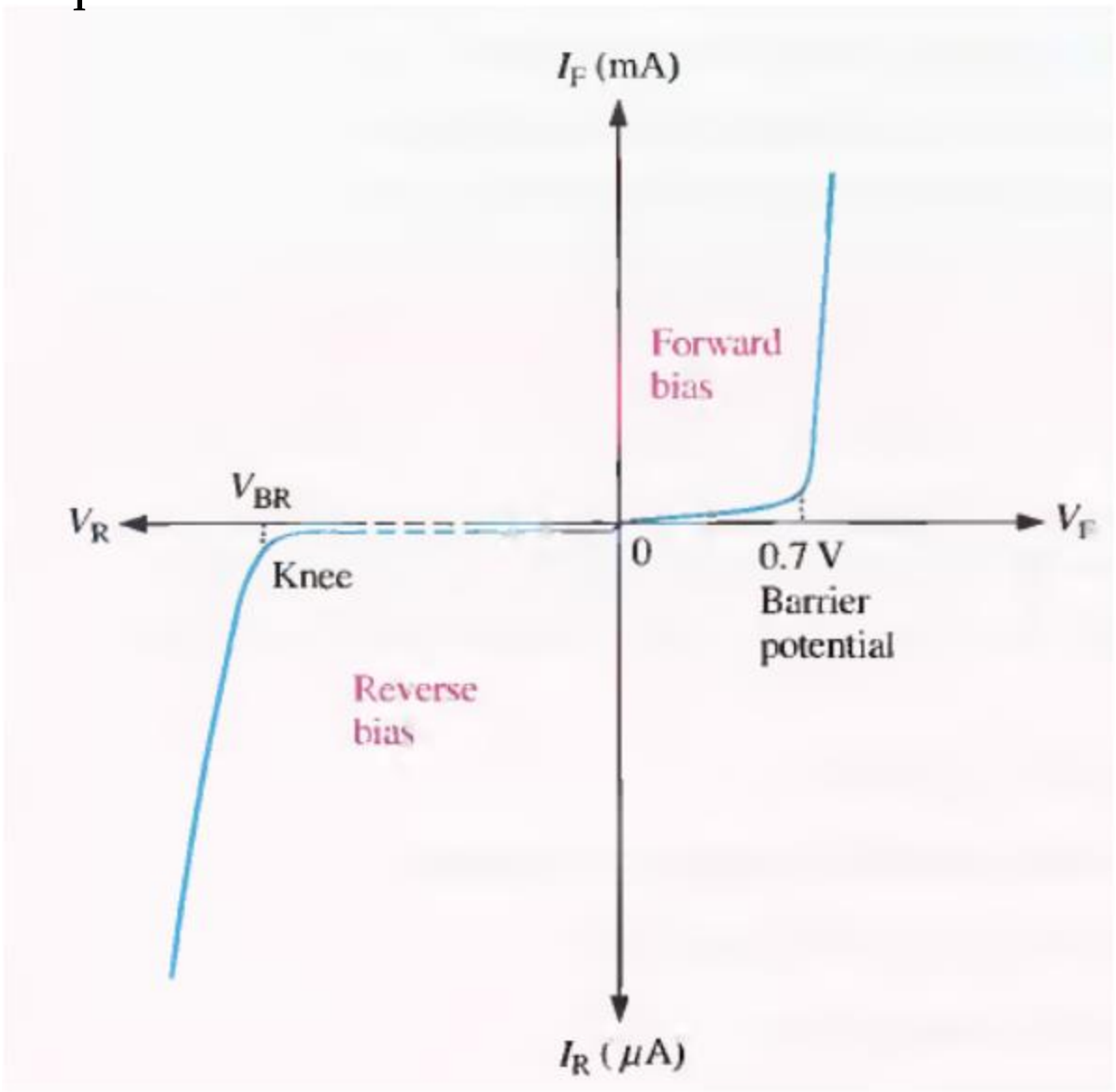


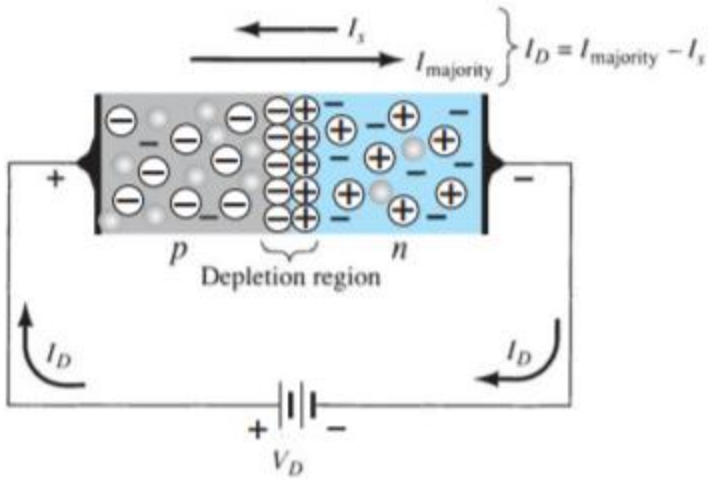
Fig. breakdown region

Avalanche breakdown (high kinetic energy)

- ✓ Avalanche breakdown, occurs in a junction which is being lightly doped and have wide depletion layer where the electric field is not strong to break covalent bonds.
- ✓ Instead, minority carriers collides with semiconductor atoms in the depletion region.
- ✓ Collision with valence electron=> covalent bonds are broken and EHP are generated.
- ✓ These generated charge carriers (EHP) are accelerated by electric field resulting in more collision, and further production of charge carriers.
- ✓ Lastly, low reverse resistance occur with rapid growth of Is.

Equation of static characteristics





Shockley's equation , for forward and reverse bias region

$$I_D = I_{majority} - I_S$$

$$I_D = I_{majority} - I_s$$

$$I_D = I_s e^{(\frac{v_D}{nV_T})} - I_s$$

$$I_D = I_s (e^{(\frac{v_D}{nV_T})} - 1)$$

Where

$$I_s \rightarrow reverse\ saturation\ current.$$

$$v_D \rightarrow is\ applied\ forward\ bias\ voltage.$$

$$n \rightarrow is\ ideality\ factor\ (physical\ conistruction).$$

$$n = 1, for\ Ge. \quad n = 2, for\ si.$$

$$v_T \rightarrow thermal\ voltage\ \&\ determined\ by$$

$$v_T = \frac{kT_{\circ k}}{q}$$

$$where$$

$$k \rightarrow is\ Boltzman's conistant\ 1.38 \times 10^{-23} \frac{J}{\circ k}$$

$$T_k \rightarrow crystal\ temperature\ in\ \circ k.$$

$$q \rightarrow is\ magnitude\ of\ electronic\ charge$$

$$q = 1.602 \times 10^{-19} c$$

Ex1-

At room temp. (27 ̊c), determine thermal voltage v_T.

Ex2-

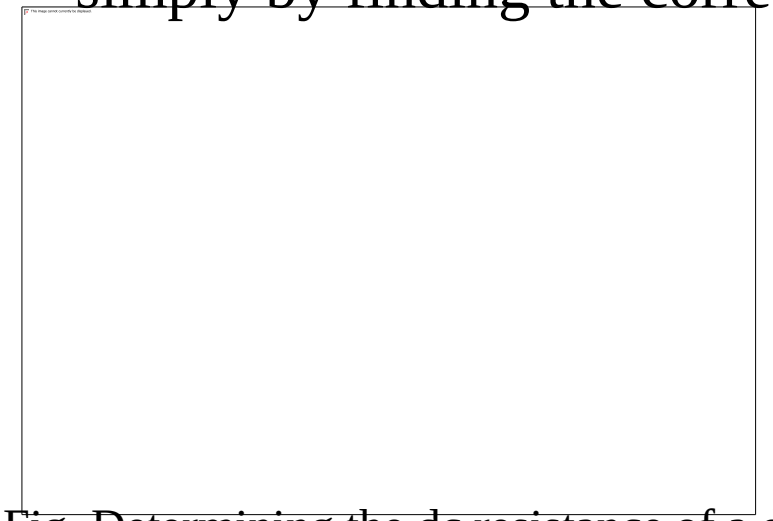
Find the change in forward bias voltage for doubling the forward current of germanium semiconductor at 290 ̊k.

Resistance Levels

- ✓ As the operating point of a diode moves from one region to another the resistance of the diode will also change due to the nonlinear shape of the characteristic curve.

DC or Static Resistance

- ✓ The application of a dc voltage to a circuit containing a semiconductor diode will result in an operating point on the characteristic curve that will not change with time.
- ✓ The resistance of the diode at the operating point can be found simply by finding the corresponding levels of VD and ID.



$$R_D = \frac{v_D}{I_D}$$

The higher the current through a diode, the lower is the dc resistance level.

Fig. Determining the dc resistance of a diode at a particular operating point

AC or Dynamic Resistance (r_d)

- ✓ The dc resistance of a diode is independent of the shape of the characteristic in the region surrounding the point of interest.
- ✓ If a sinusoidal rather than a dc input is applied, the situation will change completely.
- ✓ The varying input will move the instantaneous operating point up and down a region of the characteristics and thus defines a specific change in current and voltage.

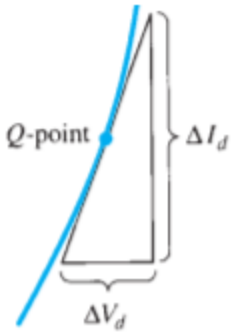


fig.a Determining the ac resistance at a Q-point.

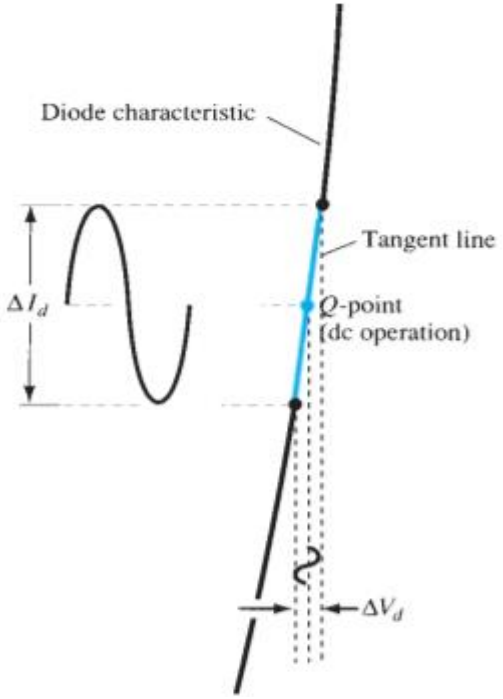


Fig.b Defining the dynamic or ac resistance.

- ✓ A straight line drawn tangent to the curve through the Q -point as shown in Fig. above define a particular change in voltage and current that can be used to determine the ac or dynamic resistance for this region of the diode characteristics.

$$r_d = \frac{\Delta v_D}{\Delta I_D} \approx \frac{26mv}{I_D}$$

The lower the Q-point the higher the resistance.

Ex-

By using analytical expression for diode current, calculate the dynamic slope resistance of Germanium diode at 290 $°k$. When a forward bias current of

i. 10μA
ii. 5mA

Resistance Level

- ✓ For a forward biased diode, as temperature increased, the forward current increases for a given value of forward voltage.
- ✓ For a reverse-biased diode, as temperature is increased, the reverse current increases.
- ✓ The barrier potential decreases as temperature increases.

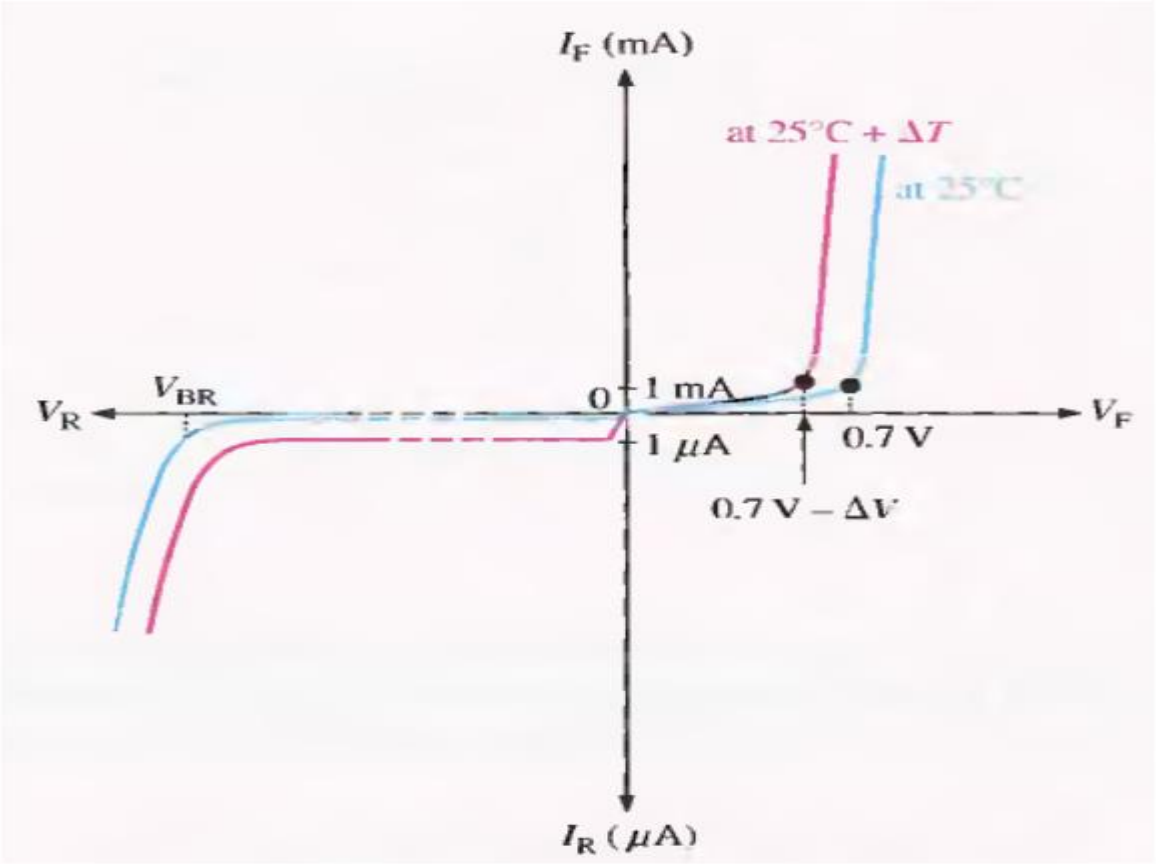


Fig. Temperature effect on VI-characteristic of diode.

Diode models

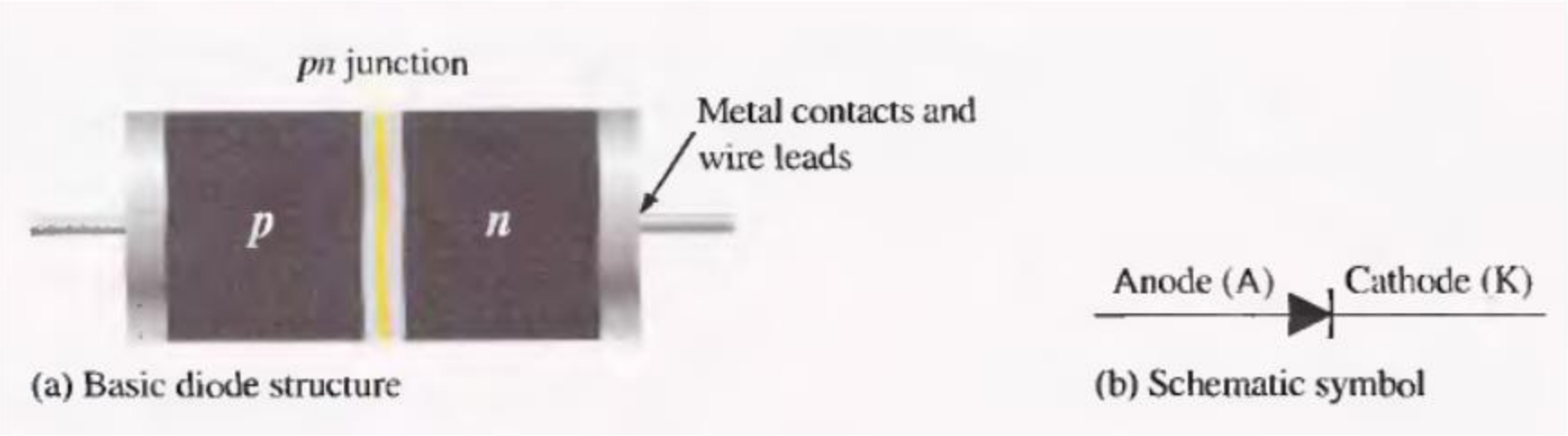


Fig. Diode structure and schematic symbol.

- ✓ The n region is called the cathode and the p region is called the anode.
- ✓ The "arrow" in the symbol points in the direction of conventional current (opposite to electron flow).

The Ideal Diode Model

- ✓ The ideal model of a diode is a simple switch.
- ✓ When the diode is forward-biased, it acts like a closed (on) switch, as shown in Figure(a).
- ✓ When the diode is reverse-biased, it acts like an open (off) switch, as shown in part (b).
- ✓ The barrier potential, the forward dynamic resistance, and the reverse current are all neglected.

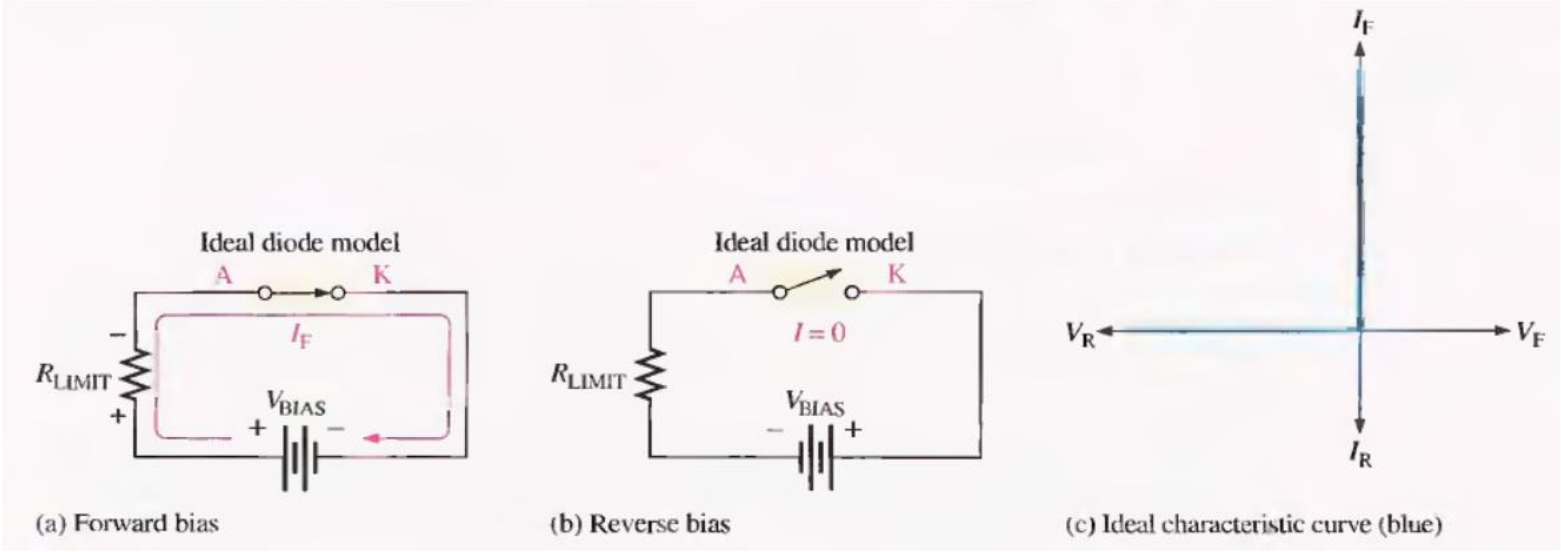


Fig. The ideal model of a diode.

Practical diode model

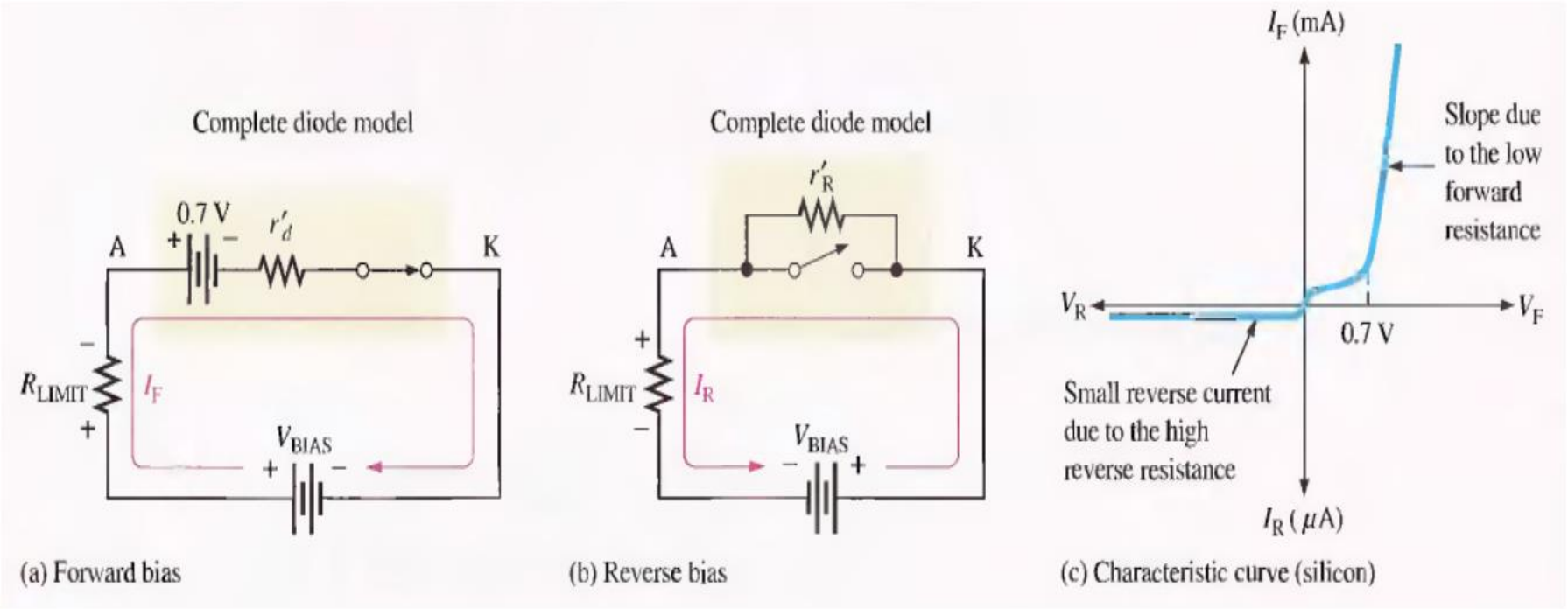


Fig. The complete model of a diode (practical).

Series diode configuration with DC input

- ✓ Diode is in the "on" state if the current established by the applied sources is such that its direction matches that of the arrow in the diode symbol, and VD = 0.7 V for silicon, VD = 0.3 V for germanium, and VD = 1.2 V for gallium arsenide.

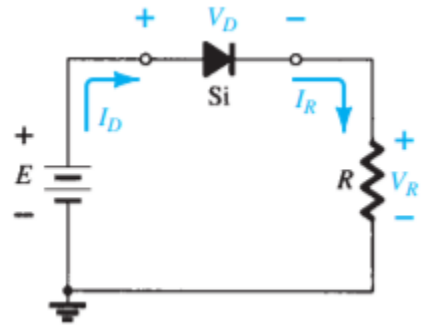


Fig. Series diode configuration.

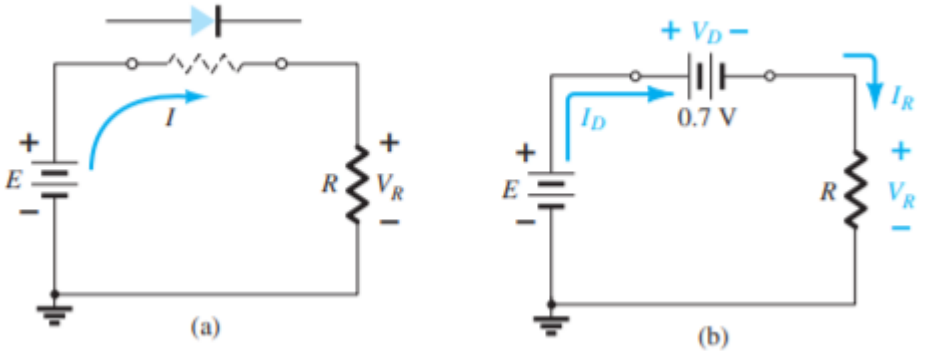


Fig. (a) Determining the state of the diode ; (b) substituting the equivalent model for the "on" diode of Fig.a .

From the series circuit above

$-E + v_D + v_R = 0$

$v_R = E - v_D$

$I_R R = E - v_D$

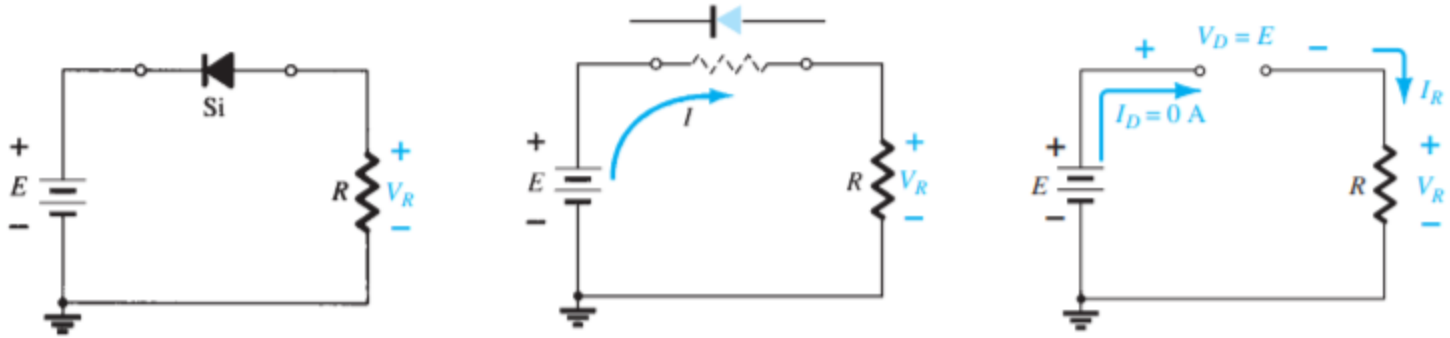
$$I_R = \frac{E - V_D}{R} = I_D$$

where, V_D=0.7v for silicon diode.

- ✓ In Fig. below the diode has been reversed.
- ✓ Mentally replacing the diode with a resistive element will reveal that the resulting current direction does not match the arrow in the diode symbol.
- ✓ The diode is in the "off" state. Due to the open circuit, the diode current is 0 A and the voltage across the resistor R is the following:
- ✓ $V_R = I_R R = I_D R = (0\ A)R = 0\ V$.

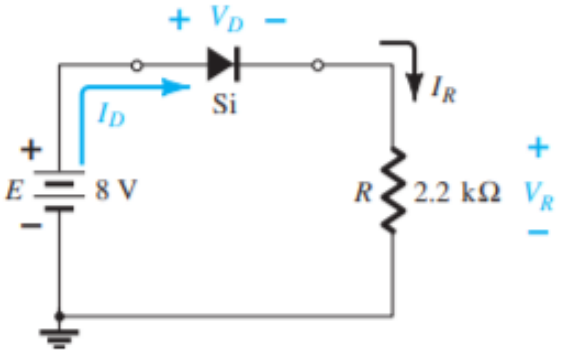
$$-E + v_D + v_R = 0$$

$$v_D = E$$



Ex.1

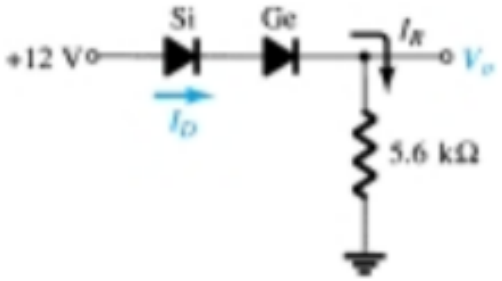
For the series diode configuration of Fig. below , determine v_D , v_R, and I_D .



- ✓ An open circuit can have any voltage across its terminals, but the current is always 0 A.
- ✓ A short circuit has a 0-V drop across its terminals, but the current is limited only by the surrounding network.

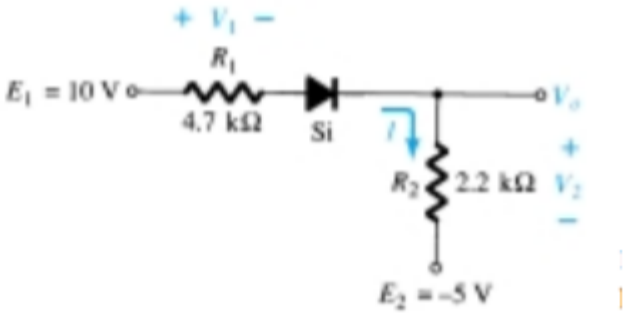
Ex-2

Determine Vo and ID for the series circuit of Fig. below



Ex-3

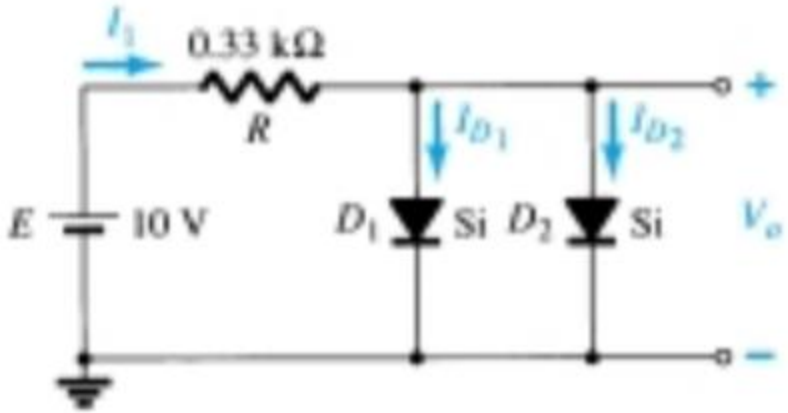
Determine I, V1, V2, and Vo for the series dc configuration of Fig. below



PARALLEL AND SERIES–PARALLEL CONFIGURATIONS

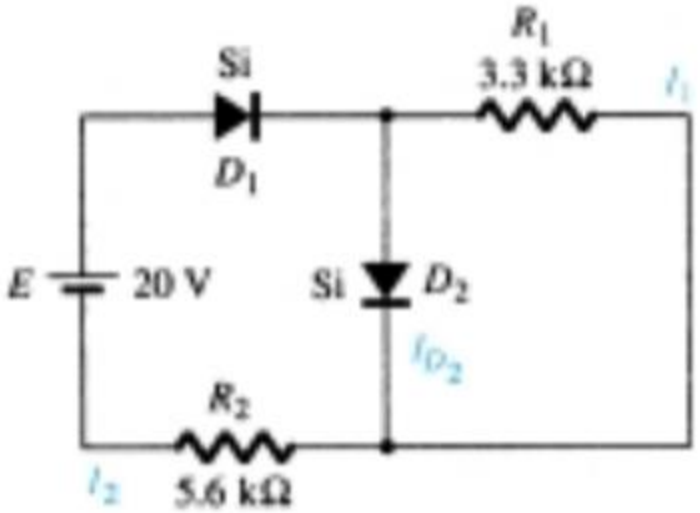
Ex.1

Determine Vo, I_1, I_{D1} , and I_{D2} for the parallel diode configuration of circuit below.



Ex-2

Determine the currents I_1, I_2 and I_{D2} for the network of circuit below



Diode applications

Rectifiers

- ✓ Rectifier is a device which is capable of converting a sinusoidal wave form in to a unidirectional waveform with a non-zero average component.

Half wave rectifier

- ✓ The process of removing one half of input signal to establish a dc level.

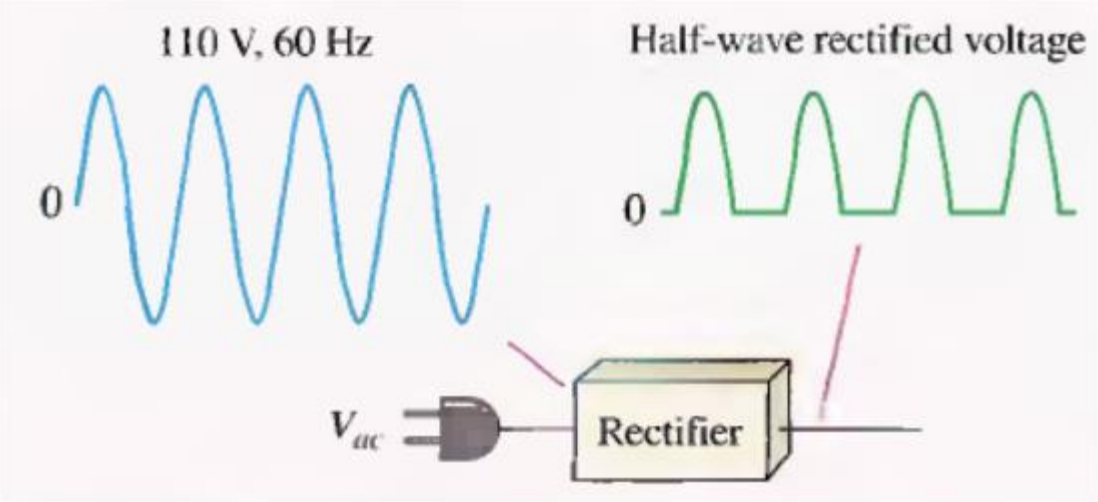


Fig. Block diagram of a rectifier.

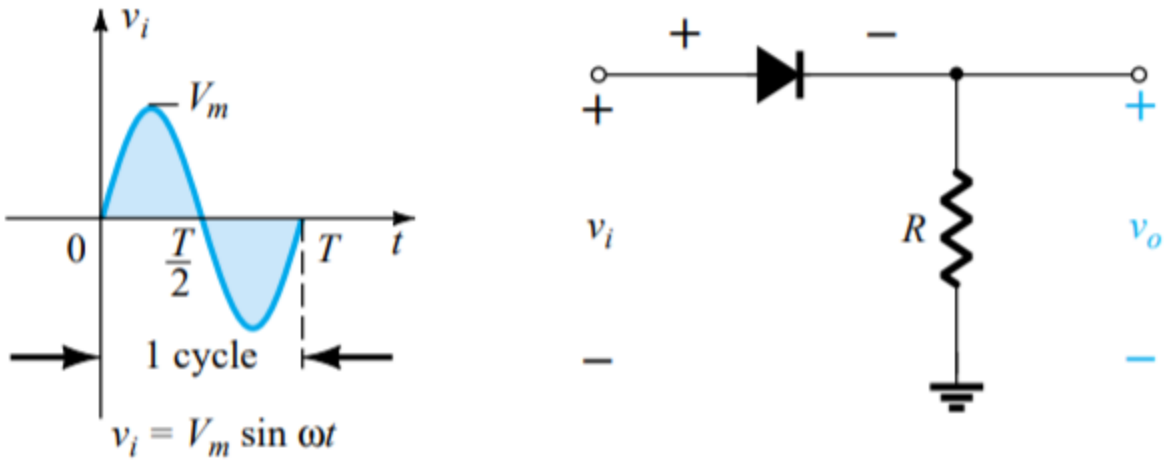


Fig. Half-wave rectifier.

- During the interval t = 0 → T/2 the output voltage looks like the positive half of the input voltage.
- During the period T/2 → T, the polarity of the input vi is as shown in Fig. below and the resulting polarity across the ideal diode produces an "off" state with an open-circuit equivalent.
- The result is the absence of a path for charge to flow and vo = iR = (0)R = 0 V for the period T/2 → T.

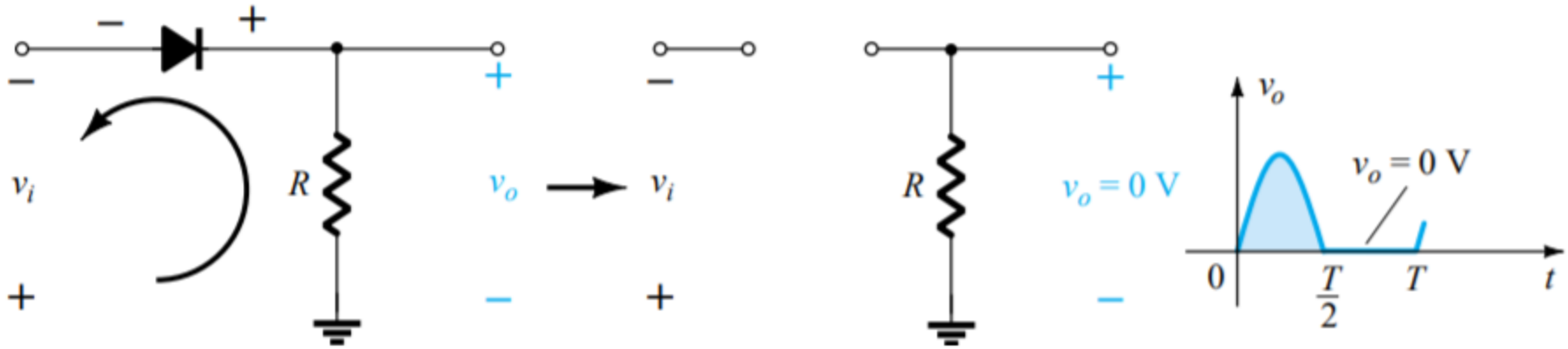


Fig. Non-conduction region (T/2 → T).

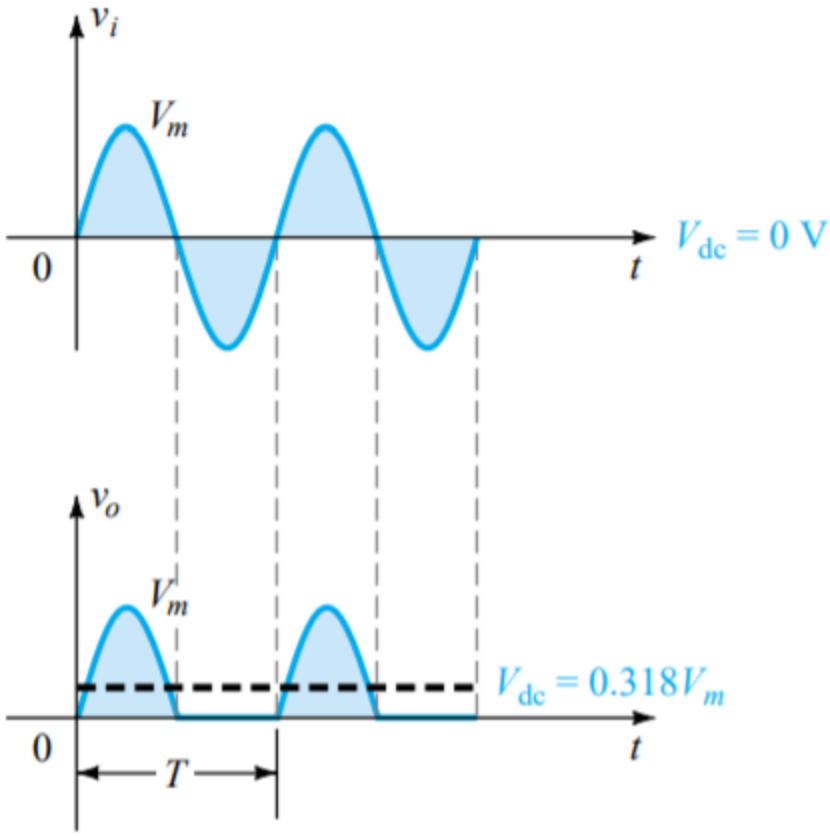


Fig. Half-wave rectified signal (vi and vo sketched together).

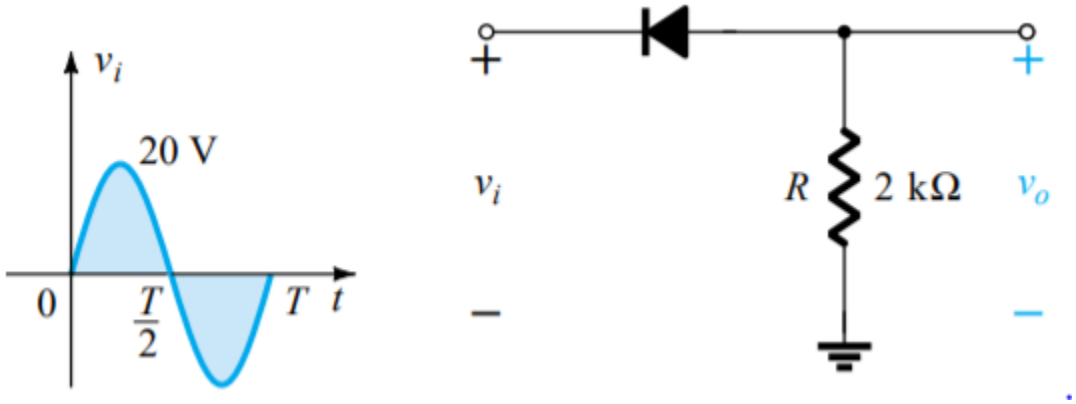
The output signal vo now has a net positive area above the axis over a full period and an average value determined by

$$V_{dc} = 0.318V_m$$

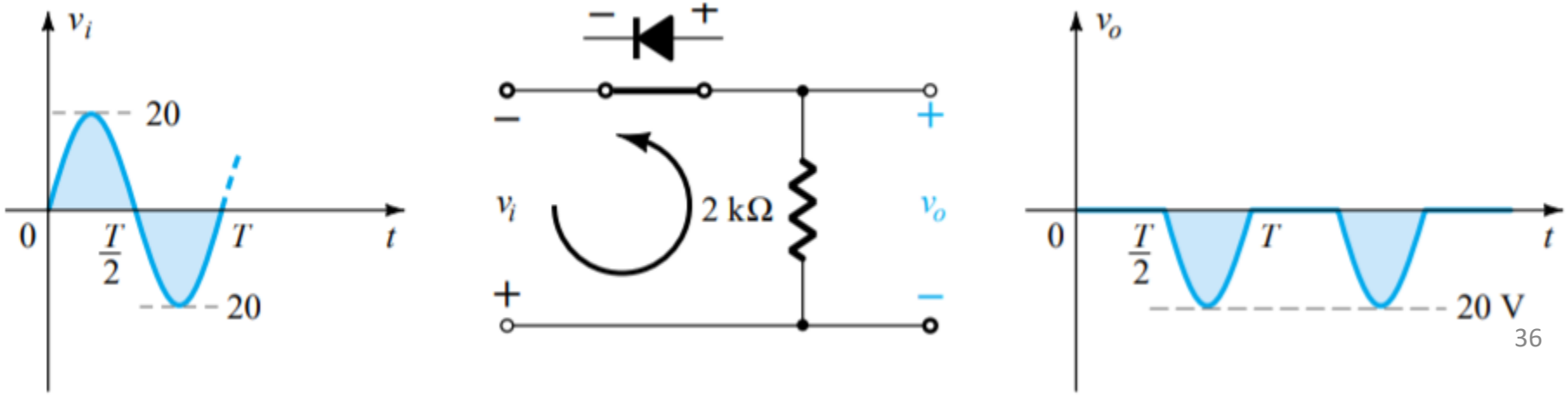
If we consider effect of barrier potential,

$$V_{dc} = 0.318(V_m - V_D)$$

For the same input if the polarity of diode is changed, the resulting output is changed.



- ✓ During the interval T=0→T/2 , the current is zero., so the output voltage is zero.
- ✓ During interval t=T/2 →T, the output voltage looks like the negative half of the input voltage.



PIV (PRV)

- ✓ Applying Kirchhoff's voltage law, it is fairly obvious that the PIV rating of the diode must equal or exceed the peak value of the applied voltage. Therefore,

$$\text{PIV rating} \geqq V_m$$ half-wave recti

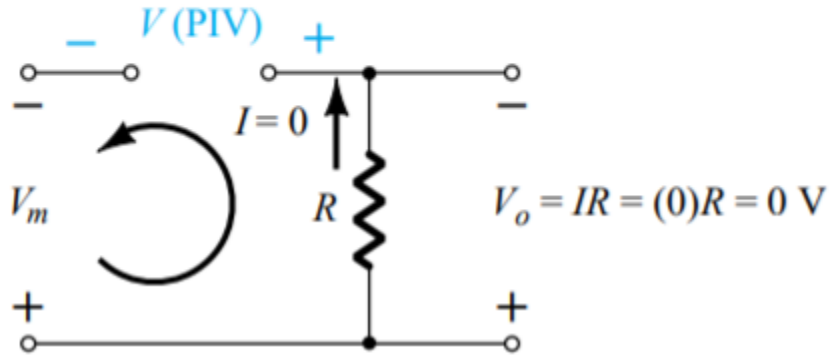
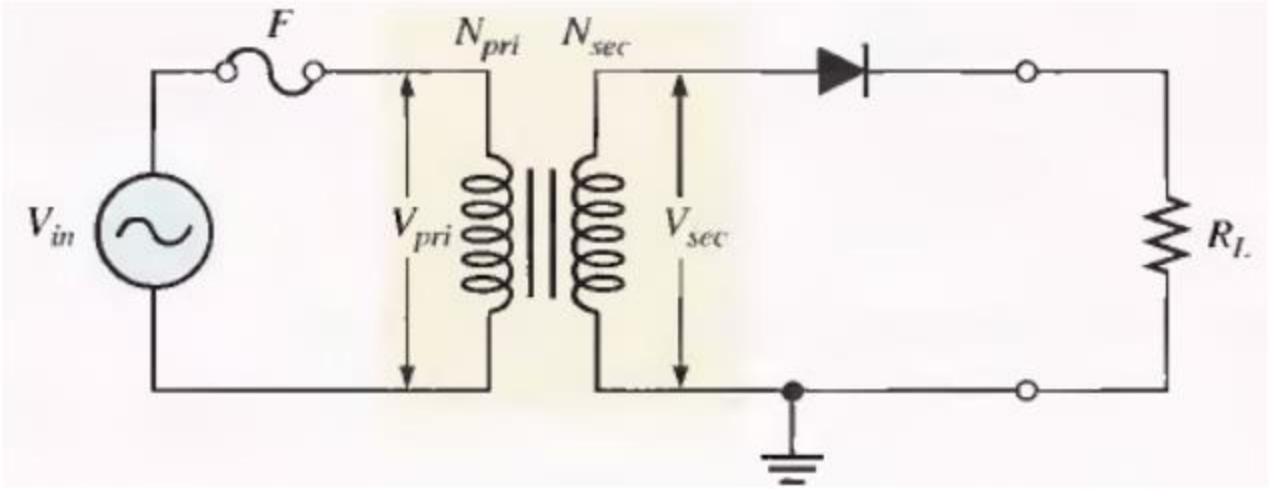


Fig. Determining the required PIV rating for the half wave rectifier.

Half-Wave Rectifier with Transformer-Coupled Input Voltage

Transformer coupling provides two advantages:-

1. It allows the source voltage to be stepped up or stepped down as needed.
2. The ac source is electrically isolated from the rectifier, thus preventing a shock hazard in the secondary.



$$turns\ ratio => n = \frac{N_{secondary}}{N_{primary}}$$

If n>1, $N_{sec} > N_{pri}$------>step up transformer

n<1, $N_{sec} < N_{pri}$------>step down transformer

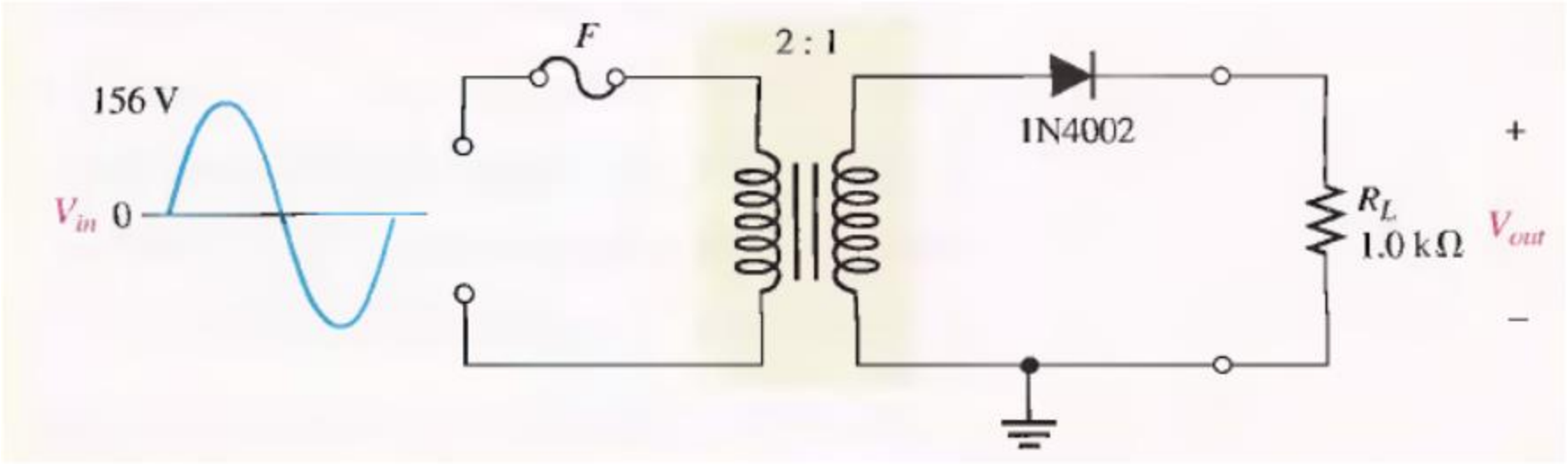
$$V_{secondary} = nV_{primary}$$

If $N_{secondary} = N_{primary}$=======>$V_{secondary} = V_{primary}$

V_p(out) = $V_P(sec) - 0.7v$ ——————— $-for\ si\ diode$

Ex-

Determine the peak value of the output voltage for Figure below if the turns ratio is 0.5.



Full wave rectifiers

- ✓ A full wave rectifier allows unidirectional (one way) current through the load during the entire 360˚ of the input cycle whereas a half wave rectifier allows current through the load only during one half of the cycle.

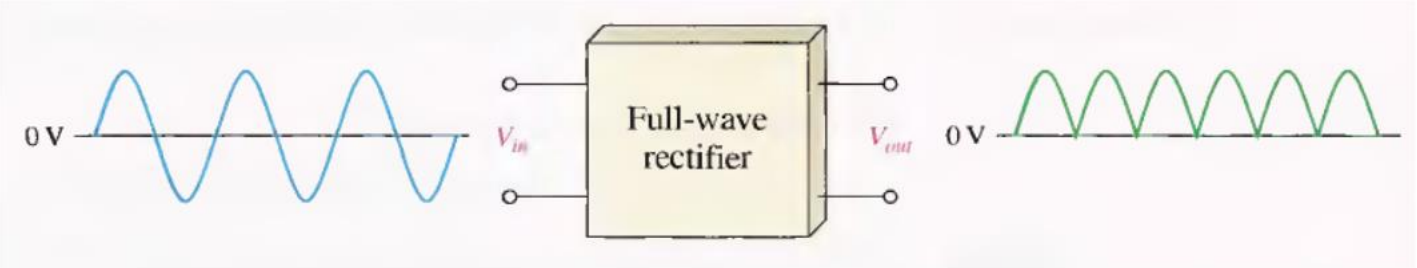


Fig. Full-wave rectification.

- ✓ The number of positive alternations that make up the full wave rectified voltage is twice that of the half wave voltage for the same time interval.

$$v_{avg} = 2 * (0.318 v_m) = 0.636 v_m$$

Bridge full-wave rectifier

- ✓ Is the most familiar full-wave rectifier.
- ✓ Four diodes are arranged in the form of bridge.

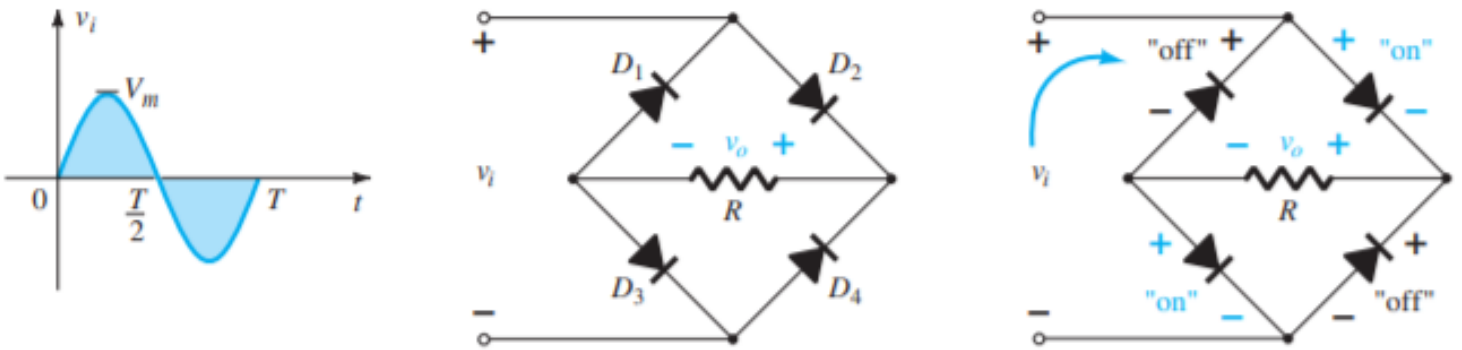


Fig. Full-wave bridge rectifier & Network for the period $0 \rightarrow \frac{T}{2}$ of the input voltage vi .

During the first half cycle:- (t=0→$\frac{T}{2}$)

- ✓ The input voltage, the upper end is positive with respect to the lower end.
- ✓ During this cycle diode D_2 & D_3 are forward biased and current flows through the load (R) from right hand side to left hand side.
- ✓ But, D_1 & D_4 are in the "off" state.
  (no current pass through D_1 & D_4, but only through D_2 & D_3)

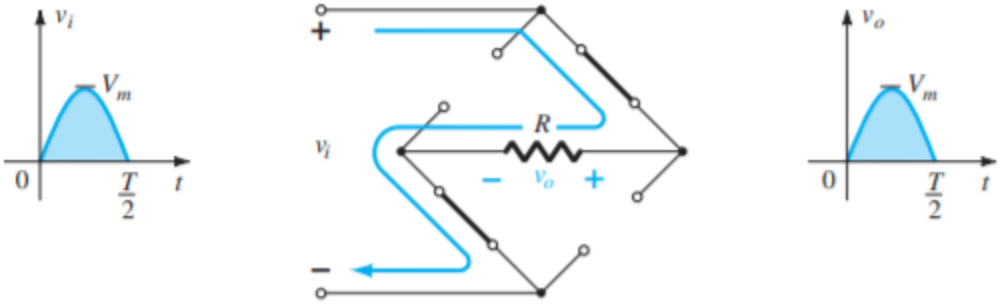


Fig. Conduction path for the positive region of vi .

During the second half cycle:- (t= $\frac{T}{2} \rightarrow$ T)

- ✓ The input voltage, the lower end is positive with respect to the upper end.
- ✓ During this cycle diode D_4 & D_1 are forward biased and current flows through the load (R) from right hand side to left hand side.
- ✓ But, D_2 & D_3 are in the "off" state.
  (no current pass through D_2 & D_3, but only through D_4 & D_1).

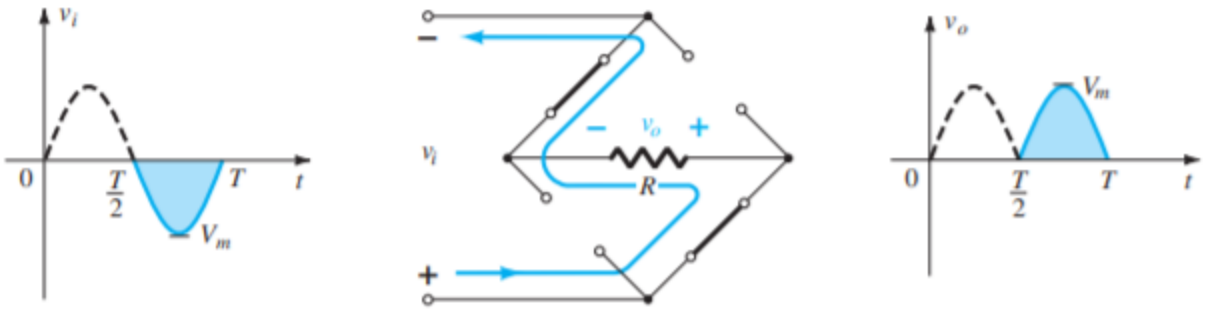


Fig. Conduction path for the negative region of vi .

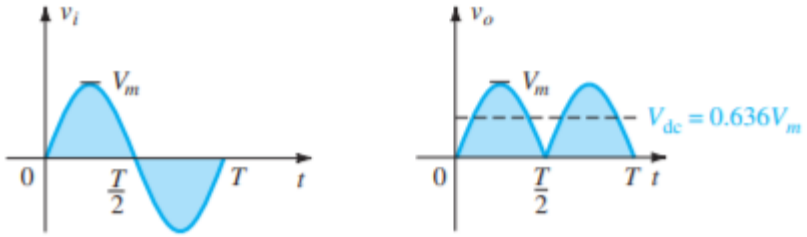


Fig. Input and output waveforms for a full-wave rectifier.

- ✓ Since the area above the axis for one full cycle is now twice that obtained for a half-wave system, the dc level has also been doubled and

$$v_{dc} = v_{avg} = 2 * (0.318v_m) = 0.636v_m$$

$$V_{dc} = 0.636\,V_m \quad \text{full-wave}$$

✓ If silicon rather than ideal diodes are employed as shown in Fig. below , the application of Kirchhoff's voltage law around the conduction path results in

$$v_i - V_K - v_o - V_K = 0$$
$$v_o = v_i - 2V_K$$

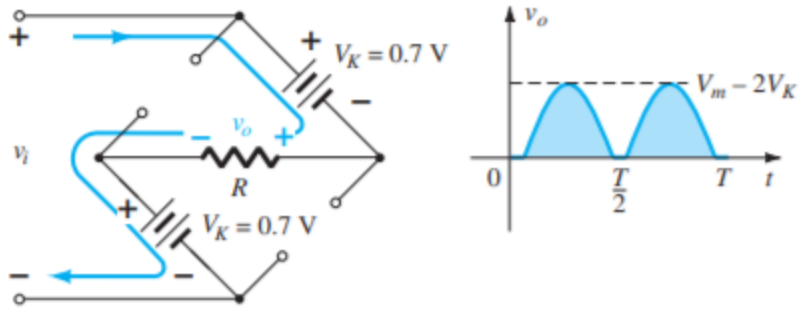


Fig. Determining $v_{o\,max}$ for silicon diodes in the bridge configuration.

The peak value of the output voltage v_o is therefore

$$v_{o\,max} = v_m - 2v_k$$

- ✓ For situations where $V_m \gg 2V_K$, the following equation can be applied for the average value with a relatively high level of accuracy:

$$V_{dc} \cong 0.636(V_m - 2V_K)$$

Peak inverse voltage (PIV)

- ✓ The required PIV of each diode (ideal) can be determined from Fig. below obtained at the peak of the positive region of the input signal.
- ✓ For the indicated loop the maximum voltage across R is Vm and the PIV rating is defined by

$$PIV \cong V_m$$

full-wave bridge rectifier

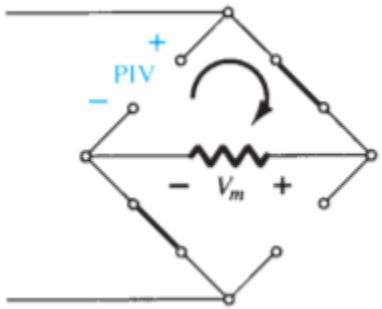


Fig. Determining the required PIV for the bridge configuration.

Center-Tapped Transformer full wave rectifier

- ✓ Only two diodes are used, requiring a center-tapped (CT) transformer to establish the input signal across each section of the secondary of the transformer.

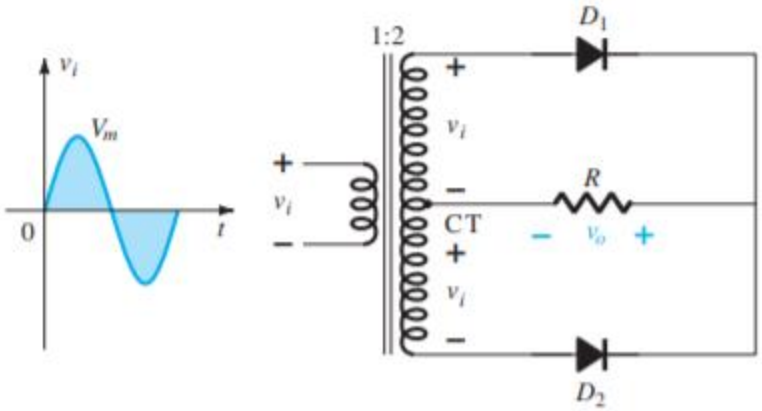


Fig. Center-tapped transformer full-wave rectifier.

- ✓ During the positive portion of vi applied to the primary of the transformer, the network will appear as shown in Fig. below with a positive pulse across each section of the secondary coil.
- ✓ D1-ON (short circuit), and D2-OFF(open circuit)

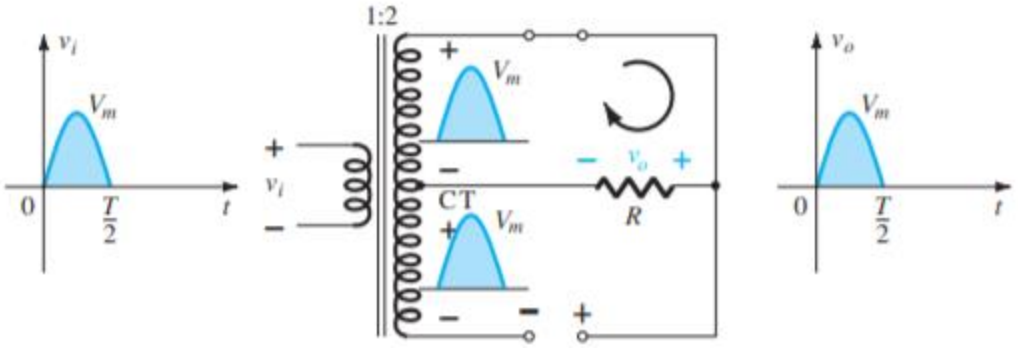


Fig. Network conditions for the positive region of vi .

- ✓ During the negative portion of the input the network appears as shown in fig below.
- ✓ D1-OFF &D2-ON

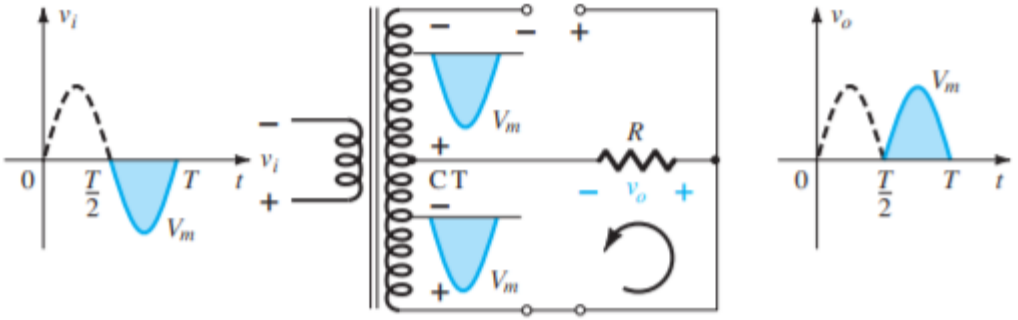
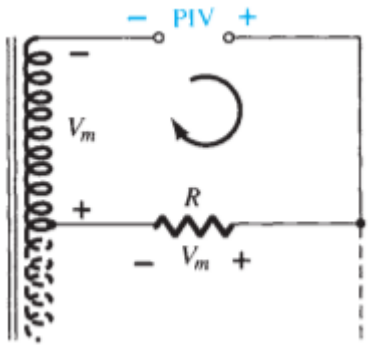


Fig. Network conditions for the negative region of vi .

PIV

PIV network of Fig. below will help us determine the net PIV for each diode for this full-wave rectifier. Inserting the maximum voltage for the secondary voltage and Vm as established by the adjoining loop results in



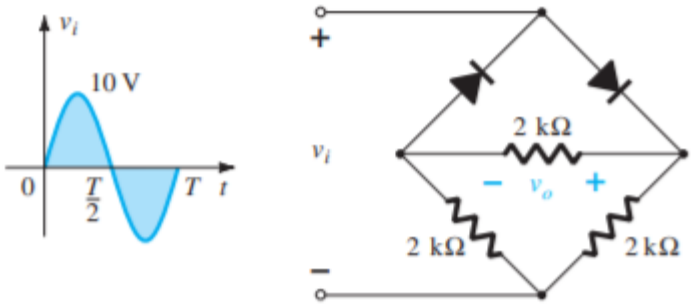
$$\text{PIV} = V_{\text{secondary}} + V_R$$
$$= V_m + V_m$$

$$\boxed{\text{PIV} \geqq 2V_m}$$ CT transformer, full-wave rectifier

Fig. Determining the PIV level for the diodes of the CT transformer full-wave rectifier.

EX-

Determine the output waveform for the network of Fig. below and calculate the output dc level and the required PIV of each diode.



Clippers

- ✓Clippers are networks that employs diode to ‘clip’ away a portion of an input signal without distorting the remaining part of the applied wave form.
- ✓Depending on the orientation of the diode, the positive or negative region of the applied signal is “clipped” off.
- ✓There are two general categories of clippers: series and parallel.
- ✓The series configuration is defined as one where the diode is in series with the load, whereas the parallel variety has the diode in a branch parallel to the load.

Series clippers

The response of the series configuration of Fig.a to a variety of alternating waveforms is provided in Fig b.

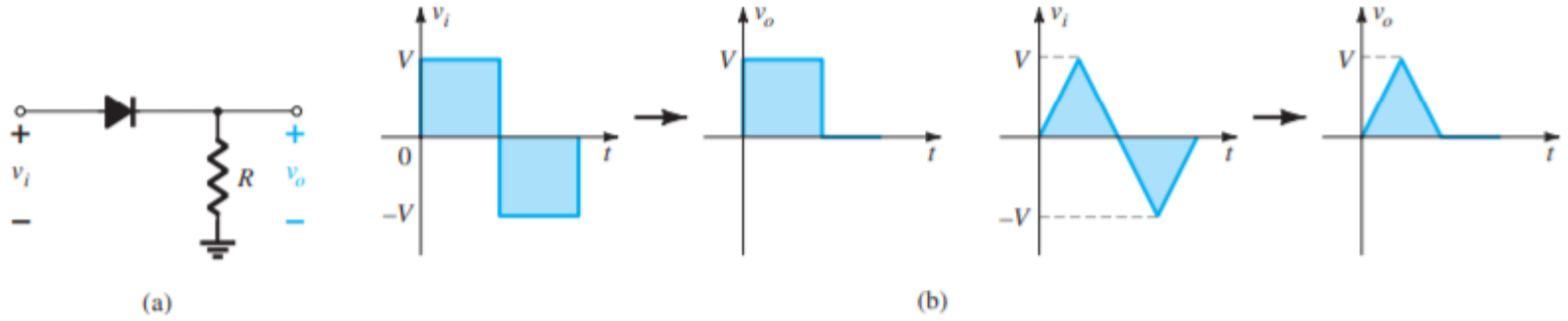


Fig. Series clipper

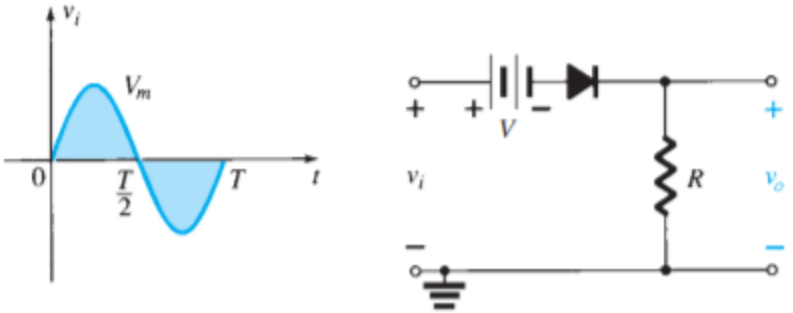


Fig. Series clipper with a dc supply.

- ✓ For the network of fig. above the diode will be on for any voltage vi that is greater than V volts and off for any lesser voltage.
- ✓ For the "off" condition, the output would be 0V due to the lack of current, and for the "on" condition it would simply be vo = vi - V as determined by Kirchhoff's voltage law.

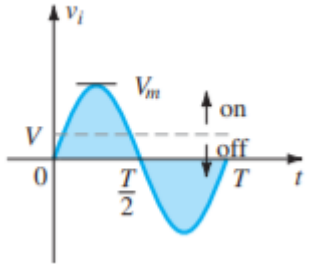


Fig. Using the transition voltage to define the "on" and "off" regions.

- ✓ For the on region if vi > V.

$$v_o = v_i - V$$

- ✓ For the "off" region, the diode is an open circuit, ID = 0 mA, and the output voltage is

$$v_o = 0\,\text{V}$$

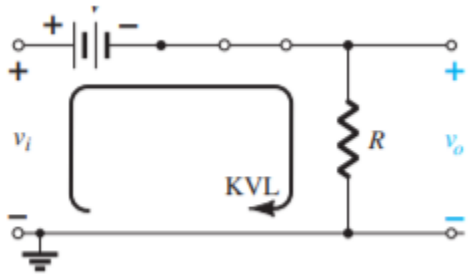


Fig. Determining vo for the diode in the "on" state.

To find the output voltage when the applied voltage has its peak value:

$$v_{o\ peak} = v_m - V$$

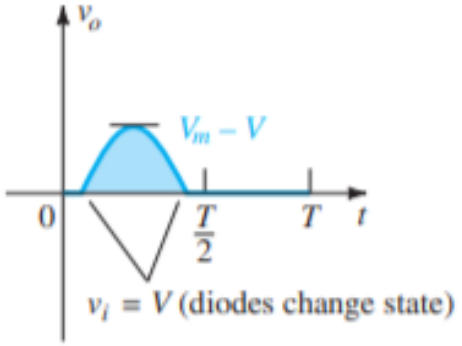
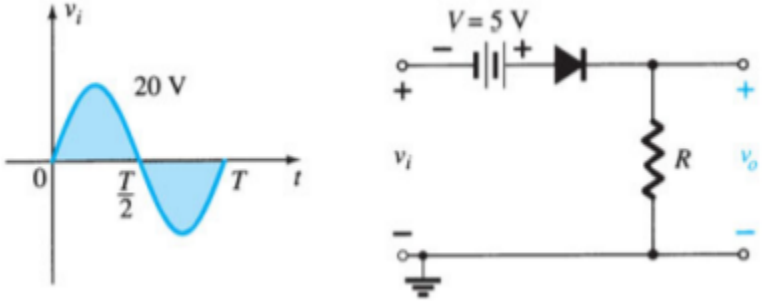


Fig. Sketching the waveform of vo using the results obtained for vo above and below the transition level.

Ex-1

Determine the output waveform for the sinusoidal input of fig below



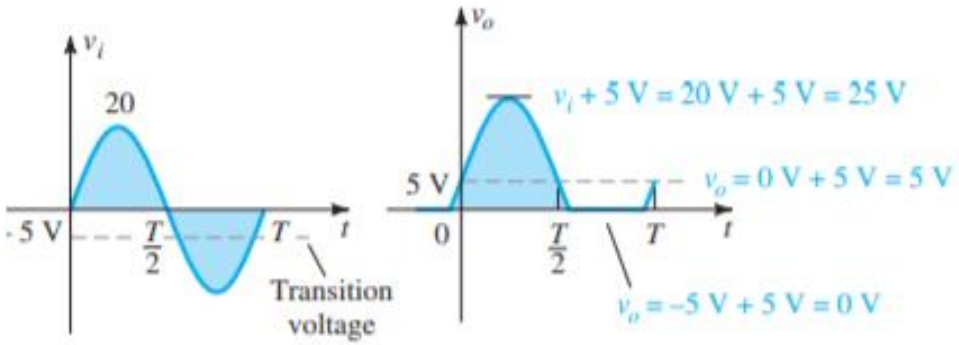
Solution

The transition state is at

vi + 5 V = 0 V or vi = -5 V

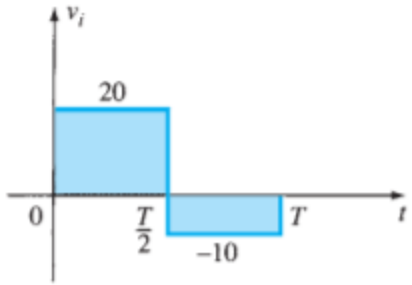
- ✓ In Fig. below horizontal line is drawn through the applied voltage at the transition level.
- ✓ For voltages less than -5 V the diode is in the open-circuit state and the output is 0 V, as shown in the sketch of vo .
- ✓ For conditions when the diode is on and the diode current is established the output voltage will be the following, as determined using Kirchhoff's voltage law:

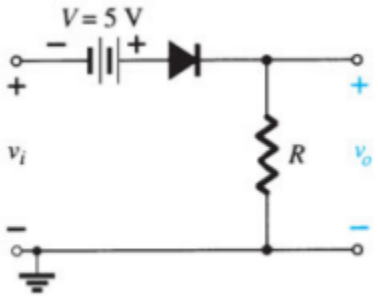
vo = vi + 5 V



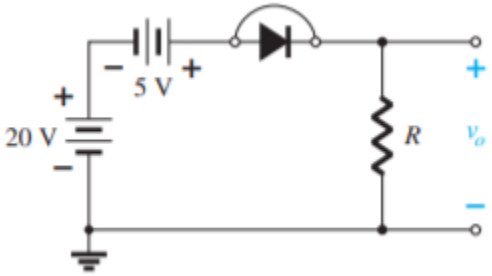
EX-2

Find the output voltage for the network examined in Ex-1 if the applied signal is the square wave of Fig. below .

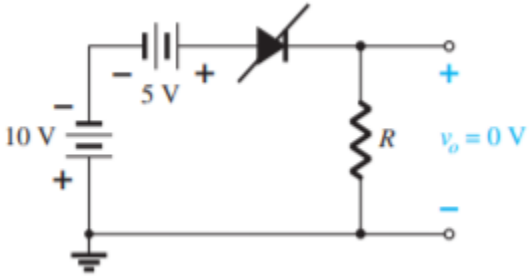




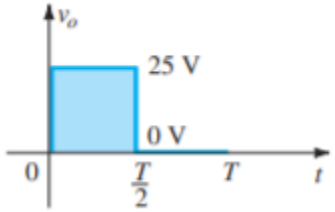
solution



v_o at $v_i = +20$ V.



v_o at $v_i = -10$ V.



Sketching v_o.

vo = 20 V + 5 V = 25 V

Parallel clippers

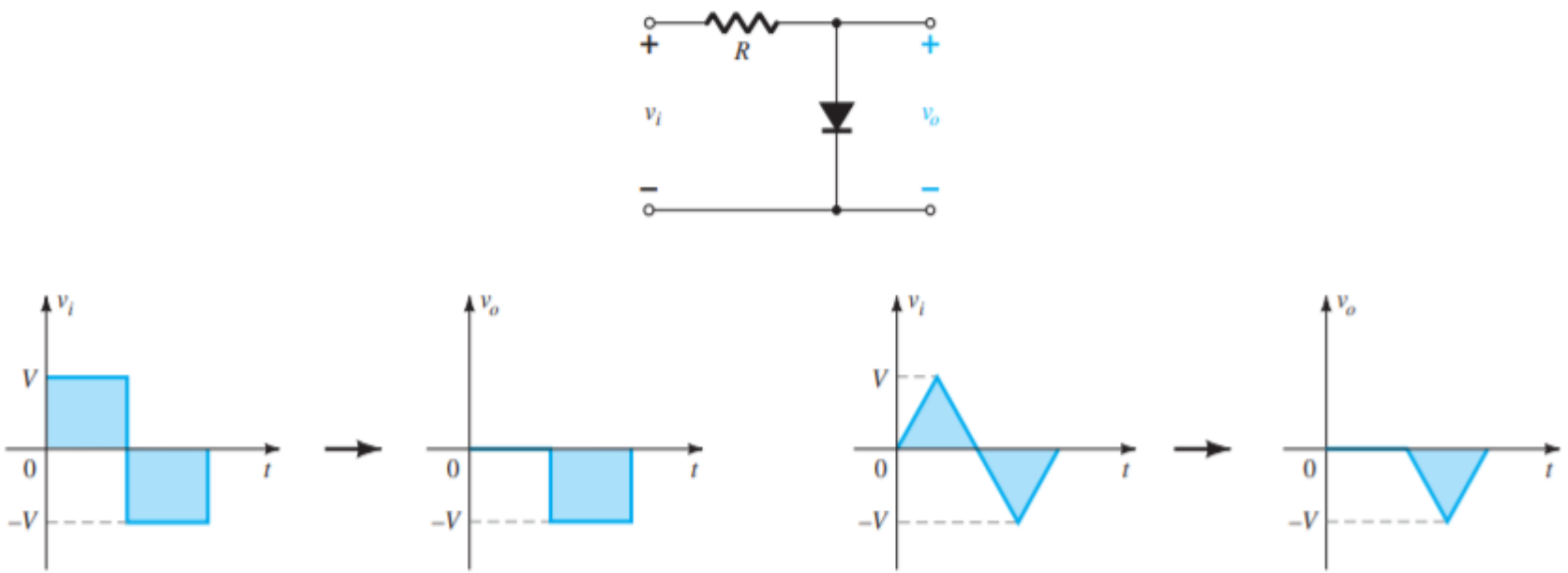
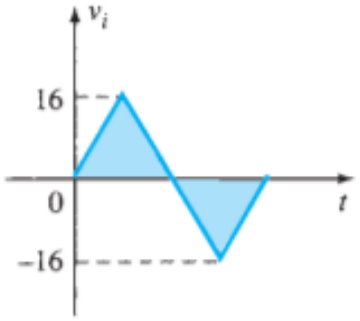
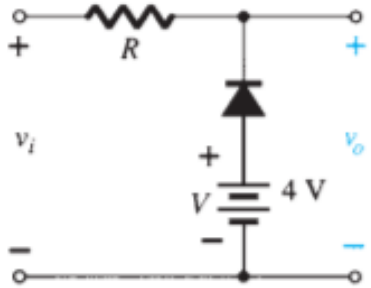


Fig. Response to a parallel clipper.

Ex-

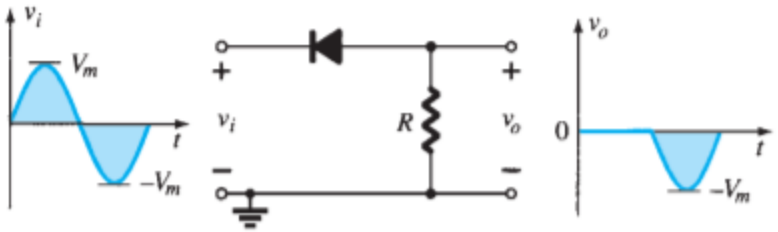
Determine vo for the network of Fig. below



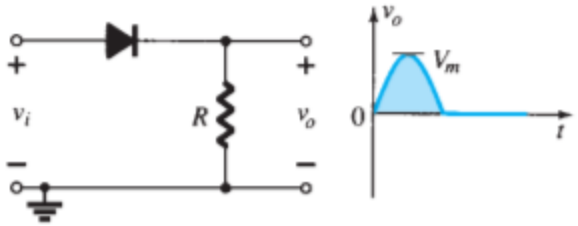


Simple Series Clippers (Ideal Diodes)

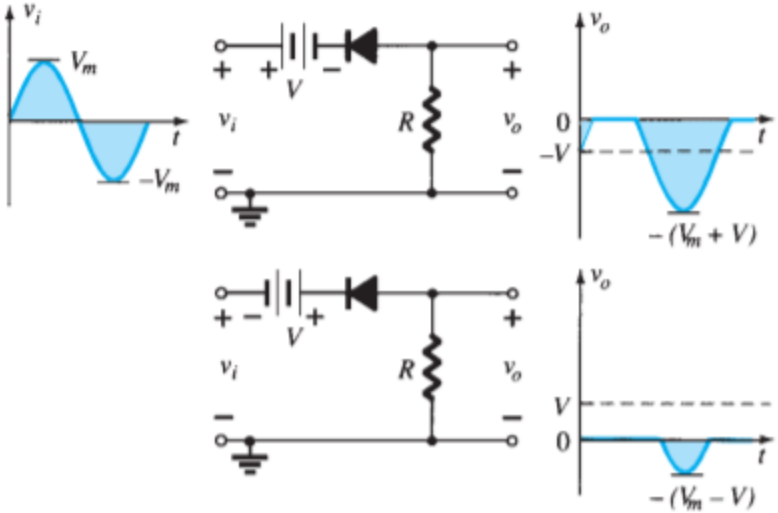
POSITIVE

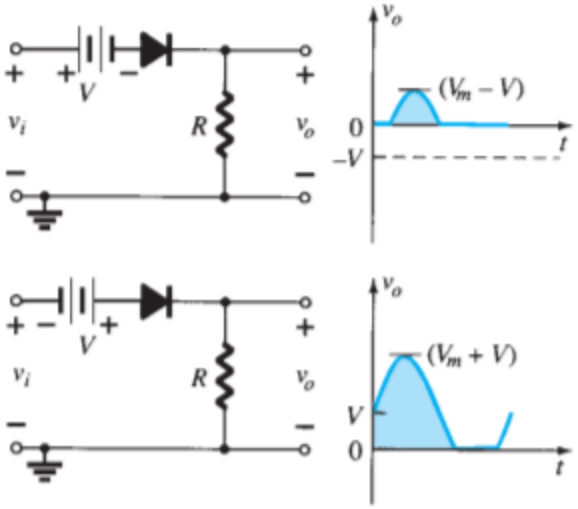


NEGATIVE

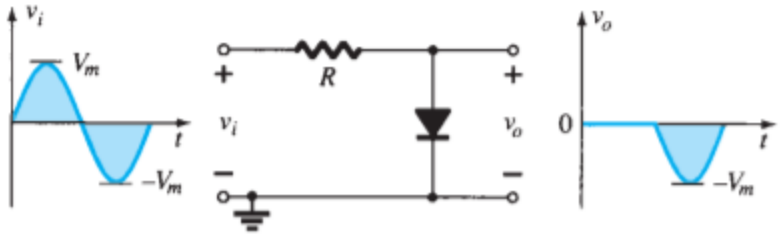


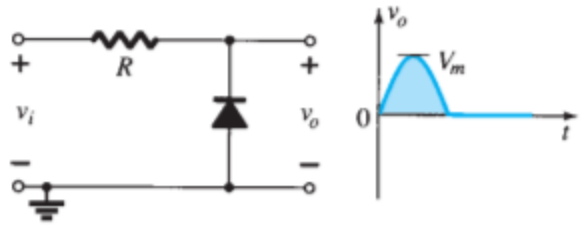
Biased Series Clippers (Ideal Diodes)



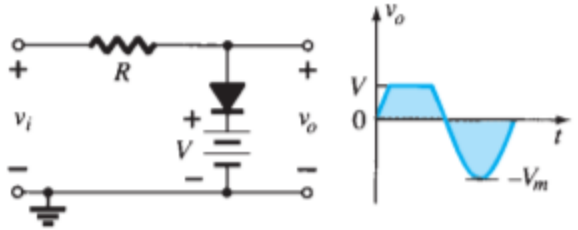


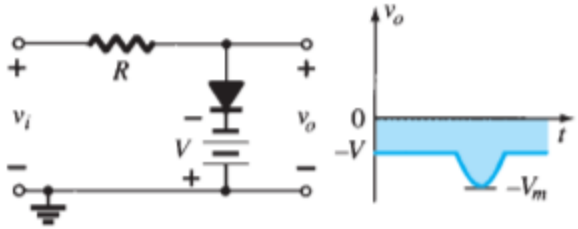
Simple Parallel Clippers (Ideal Diodes)

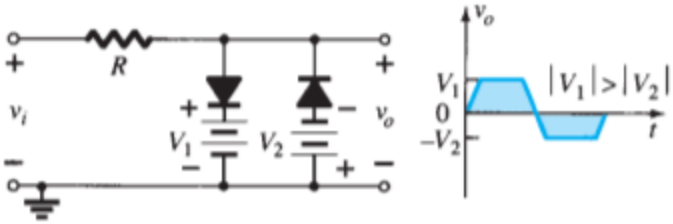


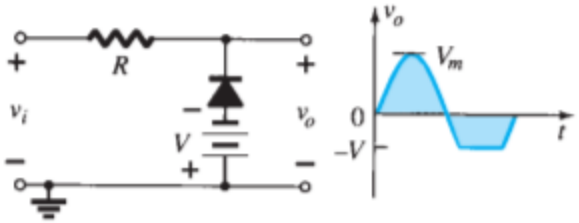


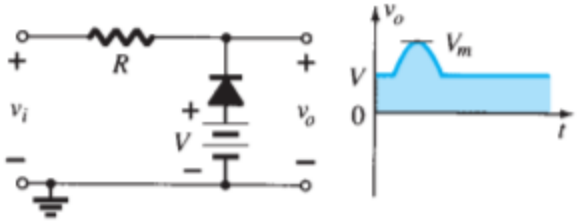
Biased Parallel Clippers (Ideal Diodes)





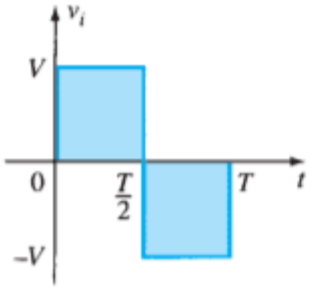


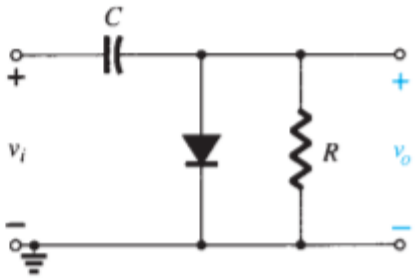




Clampers

- ✓ A clamper is a network constructed of a diode, a resistor, and a capacitor that shifts a waveform to a different dc level without changing the appearance of the applied signal.
- ✓ Clamping networks have a capacitor connected directly from input to output with a resistive element in parallel with the output signal.
- ✓ The diode is also in parallel with the output signal but may or may not have a series dc supply as an added element.

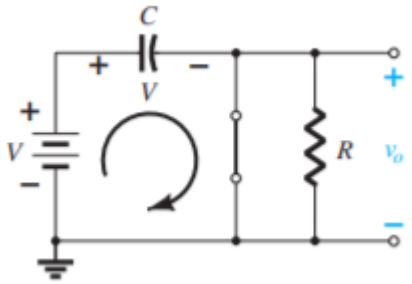




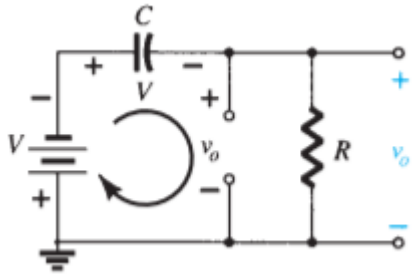
The analysis starts from forward bias

During the forward bias (t=0→T/2) (1st half cycle)

The capacitor charged to peak value of v-volts $v_c = v \; and \; v_o = 0.$



During the reverse bias (t=T/2→T) (2nd half cycle)



By using kvl

$v_c + v_i + v_D = 0$

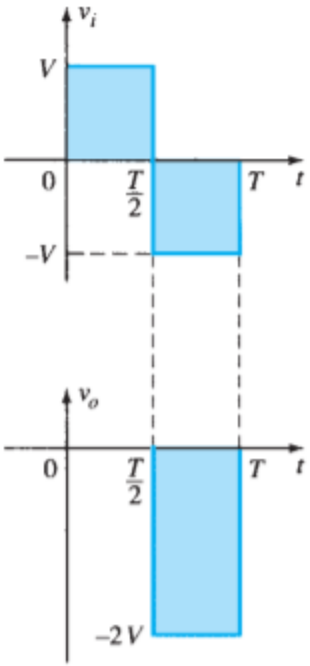
$v_D = v_c + v_i$

$v_D = V + V$

$v_D = 2V$

$v_o = -v_D$

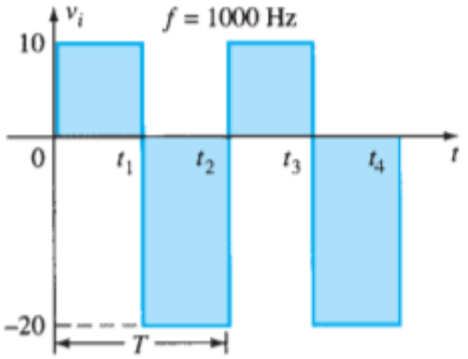
$v_o = -2V$

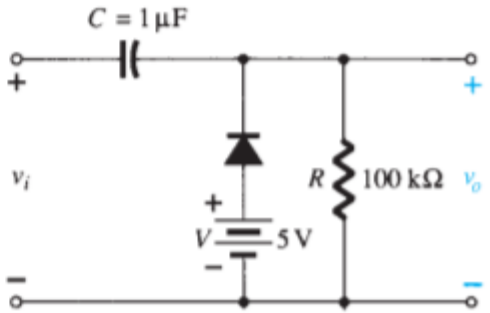


Total swing of the output matches that of the input.

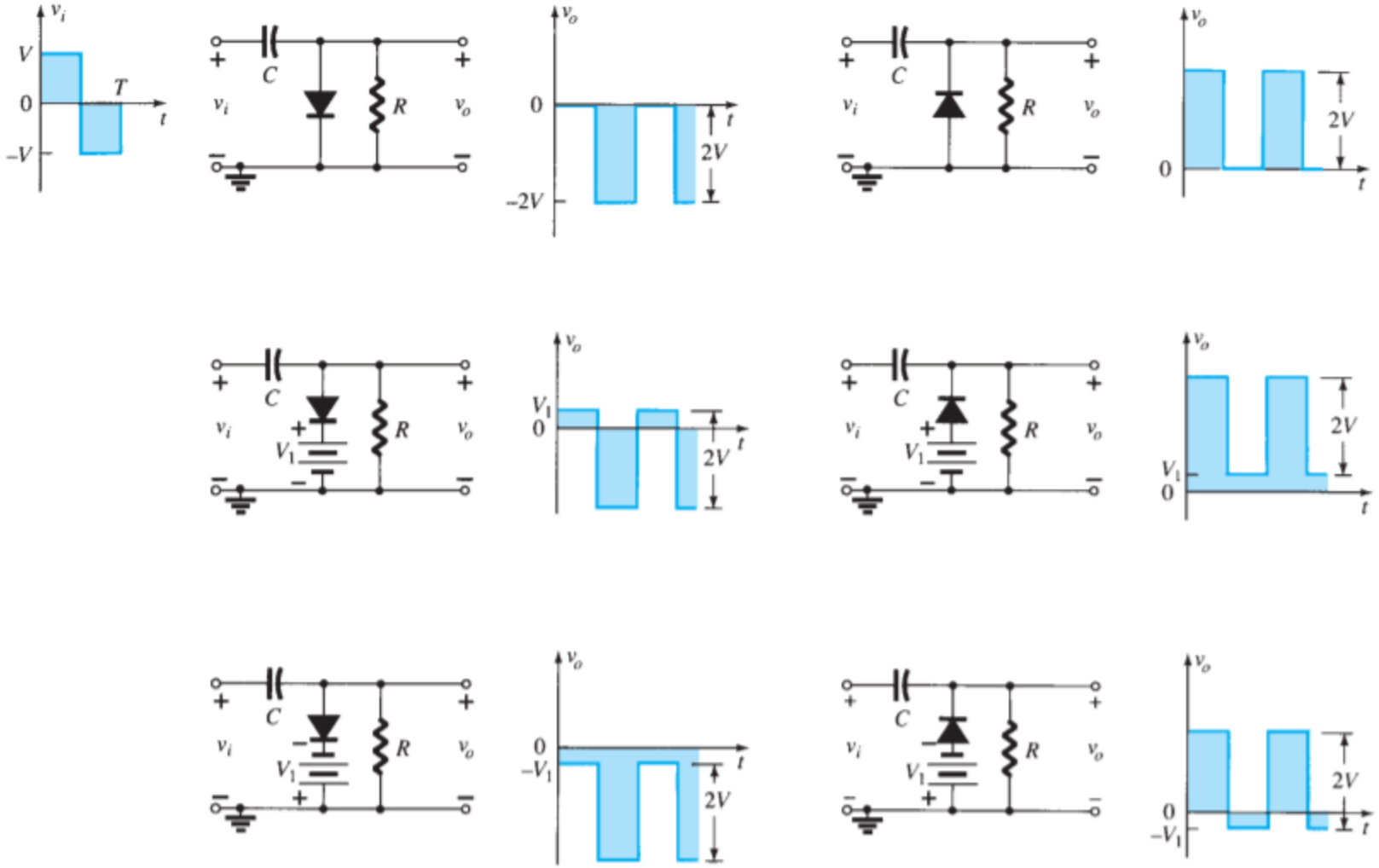
EX-

Determine vo for the network of Fig. below for the input indicated.





Clamping Networks



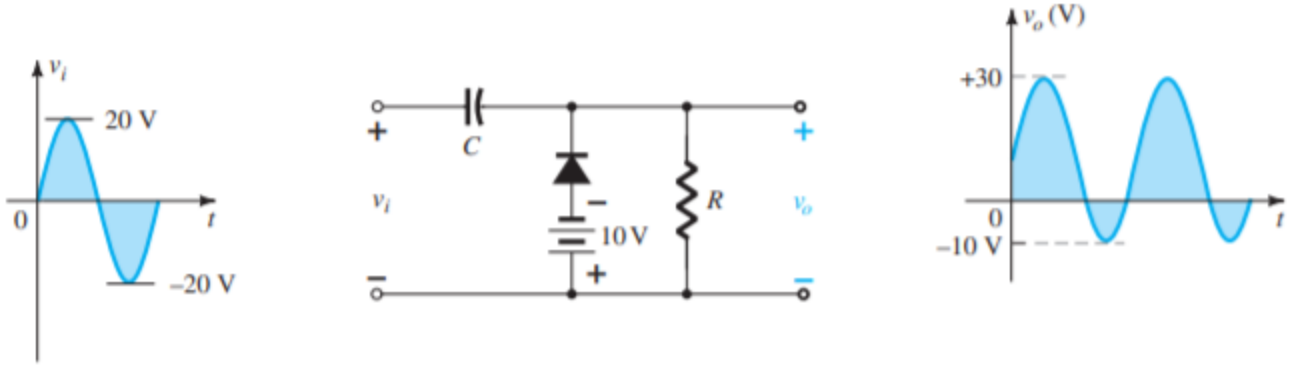


Fig. Clamping network with a sinusoidal input.

Voltage multiplier circuit

- ✓ Voltage-multiplier circuits are employed to maintain a relatively low transformer peak voltage while stepping up the peak output voltage to two, three, four, or more times the peak rectified voltage.

Voltage Doubler

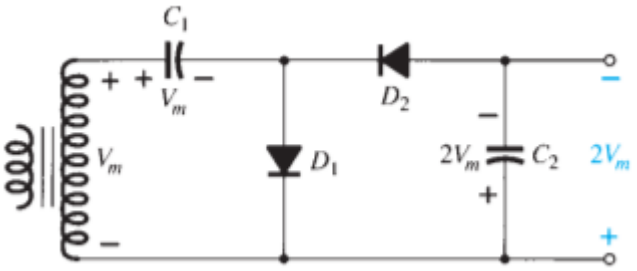


Fig. Half-wave voltage doubler.

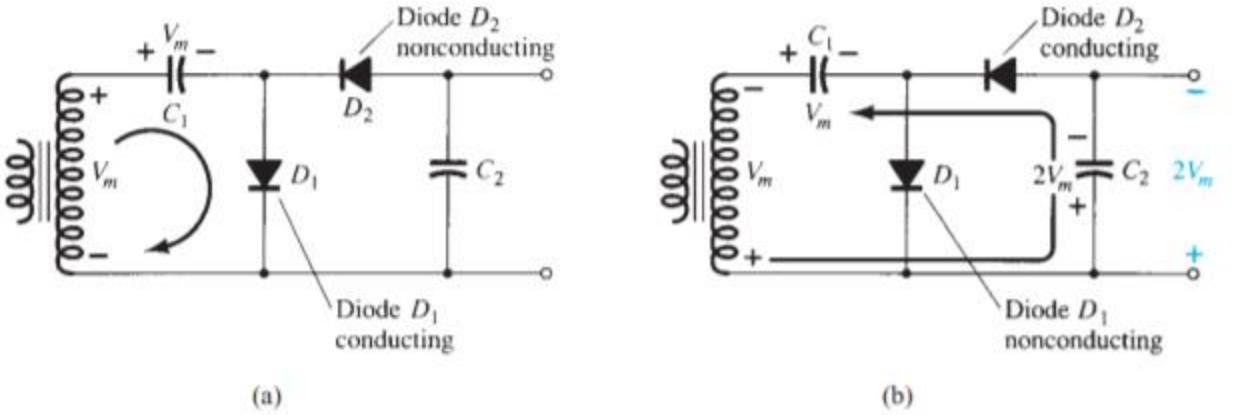


Fig. Double operation, showing each half-cycle of operation: (a) positive half-cycle; (b) negative half-cycle.

During the positive half cycle diode D1-ON

D2-OFF

Capacitor C1-is charged up to peak rectified voltage v_m.

$$v_{c1\,peak} = v_m$$

$$v_o = 0V$$

During the negative half cycle diode D1-OFF

D2-ON

Capacitor C2-is charged through D2. From circuit b).

$$-v_{c1} - v_m - v_{c2} = 0$$

$$v_{c2} = -v_{c1} - v_m$$

$$v_{c2} = -v_m - v_m$$

$$v_{c2} = -2v_m$$

$v_o = -v_{c2}$

$v_o = 2v_m$

Full wave voltage doubler

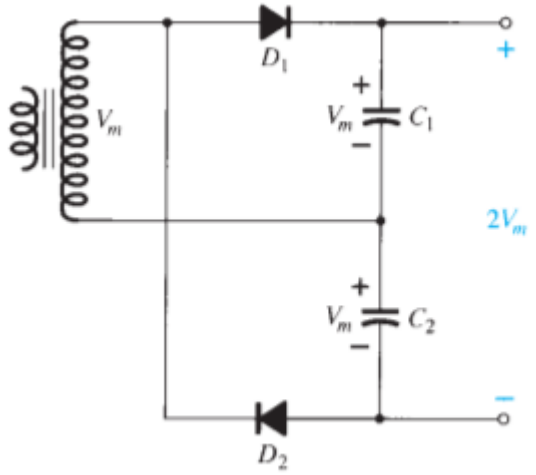


Fig. Full-wave voltage doubler.

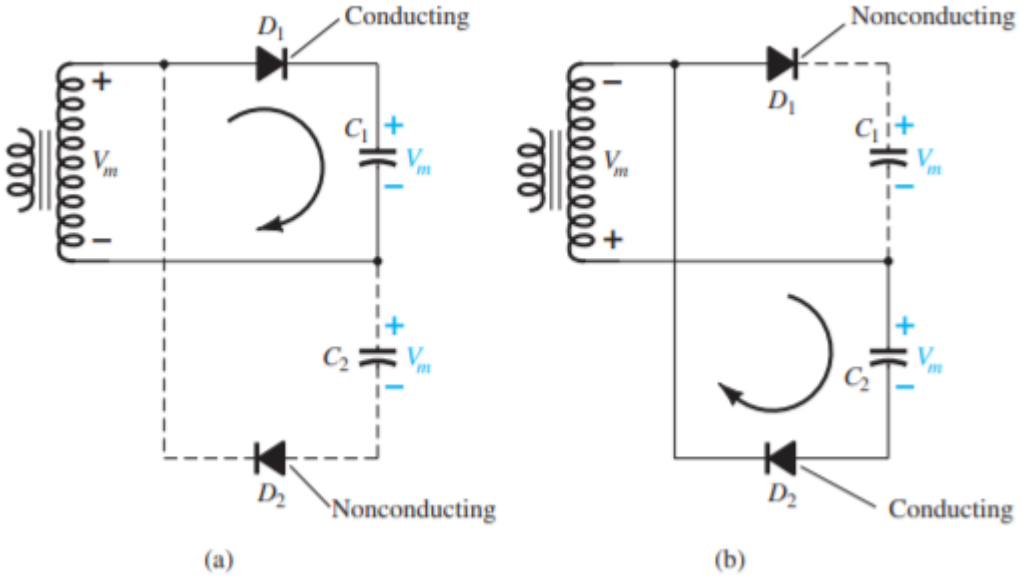


Fig. Alternate half-cycles of operation for full-wave voltage doubler.

During 1st half

- ✓ D1-conducts & D2-non conducting
- ✓ Capacitor c1-charged up to peak voltage v_m

During 2nd half cycle

- ✓ D2-conducts & D1-non conducting
- ✓ Capacitor c2-charged up to peak voltage v_m

- ❖ If no load current is drawn from the circuit, the voltage across capacitor c1 & c2 is $2v_m$.
- ❖ If load current is drawn from the circuit, the voltage across capacitor c1 &c2 is the same as that across capacitor fed by a full wave rectifier.

Voltage Tripler and Quadrupler

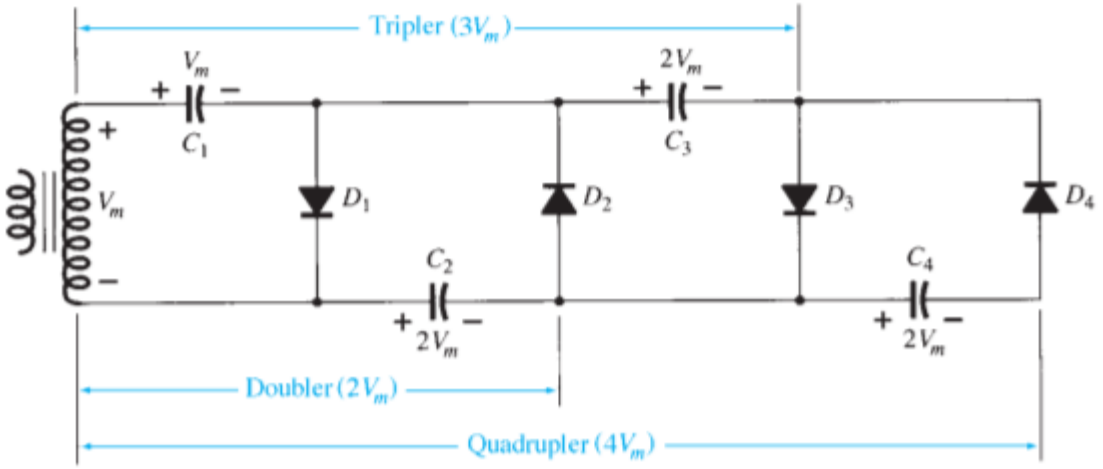


Fig. Voltage tripler and quadrupler.

1st positive half cycle

- ✓ In operation, capacitor c_1 charges through diode D_1 to a peak voltage v_m during the positive half-cycle of the transformer secondary voltage.
- ✓ Capacitor c_2 charges to twice the peak voltage, $2v_m$, developed by the sum of the voltages across capacitor c_1 and the transformer during the negative half-cycle of the transformer secondary voltage.

1st negative half cycle

- ✓ Capacitor c_2- charges to $2v_m$, developed by sum of v_{c1}+v_{sec}.

2nd positive half cycle

- ✓ Voltage across capacitor c_2 charges capacitor c_3 to the same $2v_m$ peak voltage.
- ✓ Diode D_3- conducts.

2nd negative half cycle

- ✓ In this negative half cycle –diode D_2 & D_4 conducts with capacitor c_3, charging capacitor c_4 to $2v_m$.

Voltage across capacitor $c_2 = 2v_m$
Voltage across capacitor c_1 & $c_3 = 3v_m$
Voltage across capacitor c_2 & $c_4 = 4v_m$

Power supply filters and regulators

- ✓ A power supply filter ideally eliminates the fluctuations in the output voltage of a half- wave or full-wave rectifier and produces a constant-level dc voltage.
- ✓ Filters → are smoothing circuits.

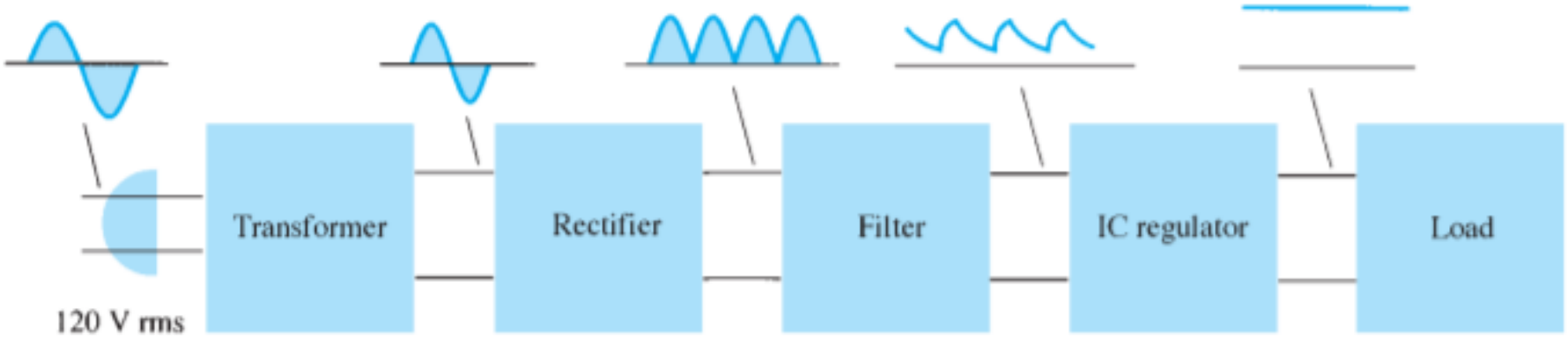


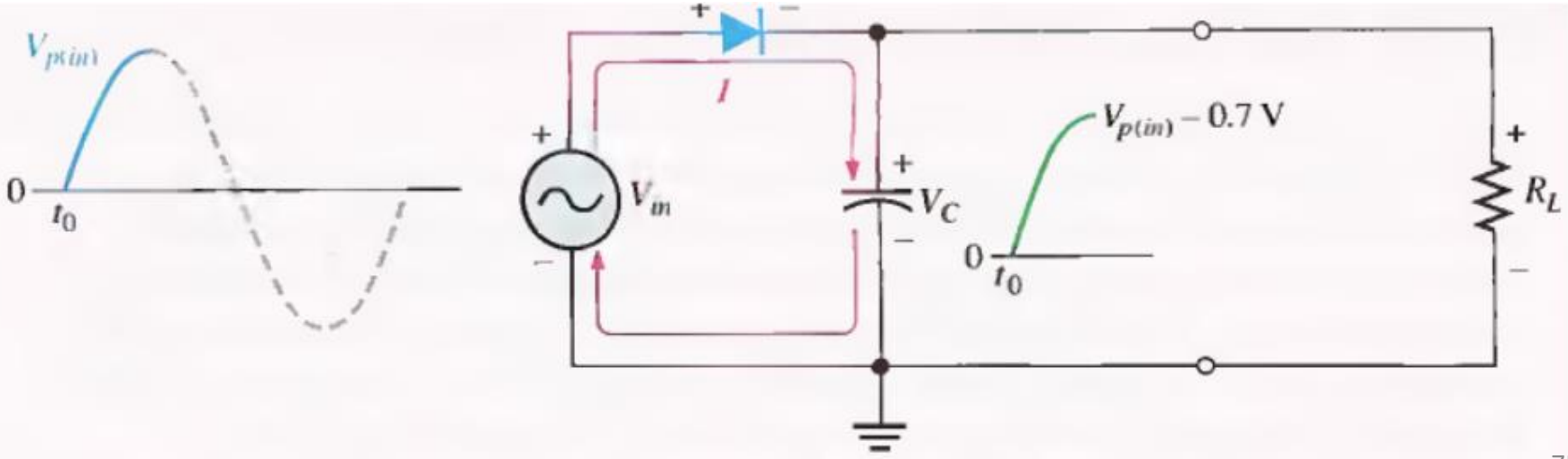
Fig. Block diagram showing parts of a power supply.

Rectifiers

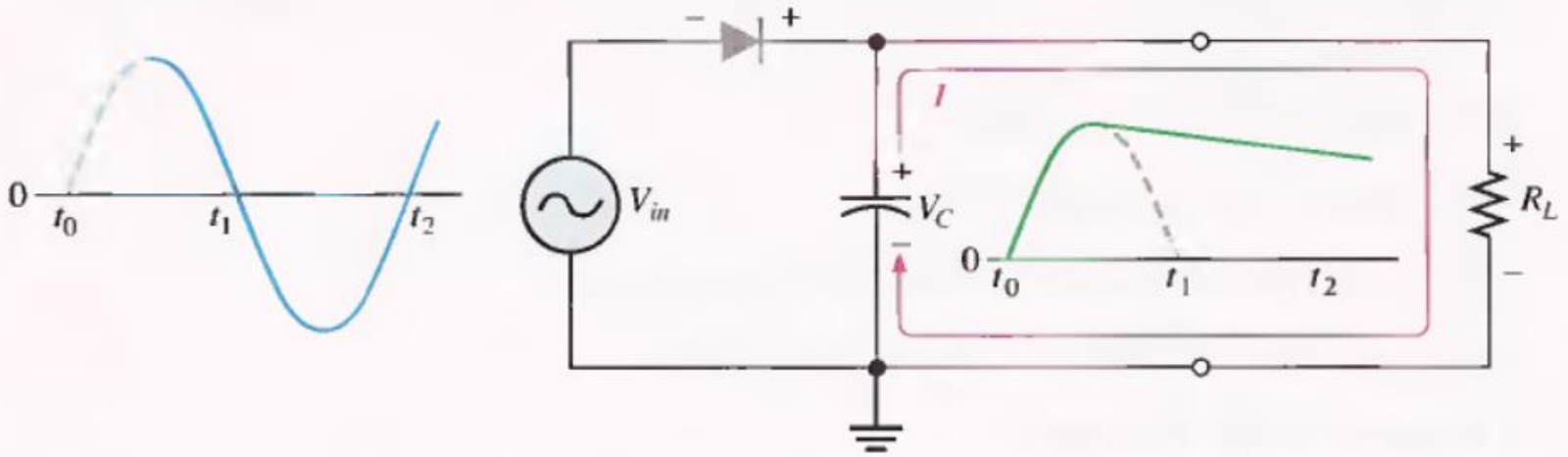
- ✓ A rectifier circuit is necessary to convert a signal having zero average value into one that has a nonzero average.
- ✓ The output resulting from a rectifier is a pulsating dc voltage and not yet suitable as a battery replacement.

Capacitor-Input filter

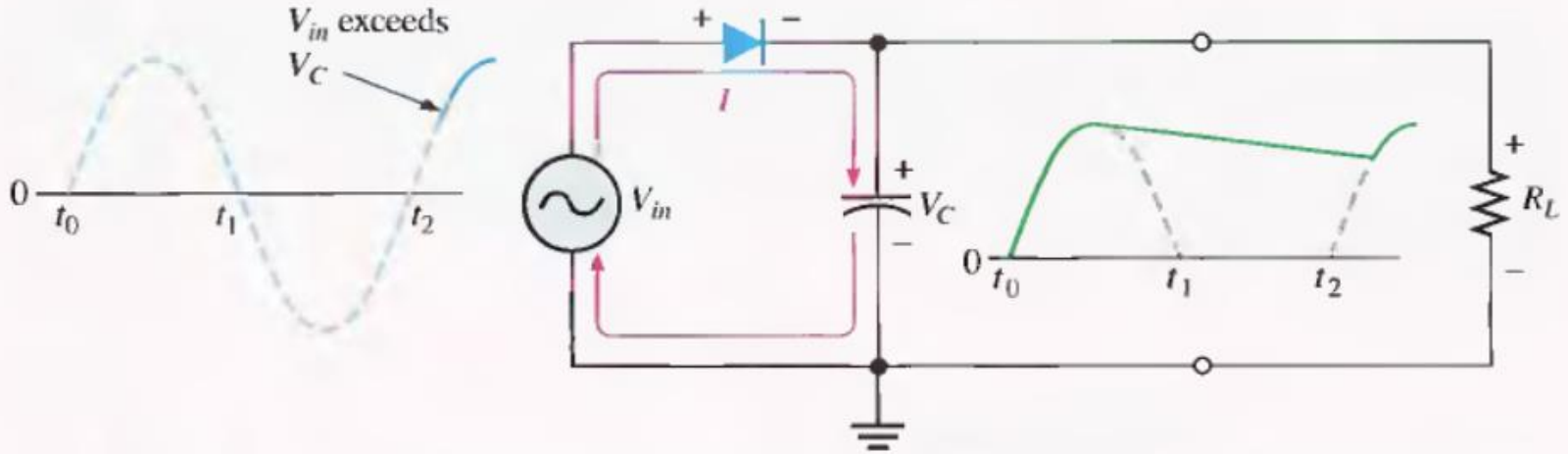
- ✓ A half-wave rectifier with a capacitor-input filter is shown below.
- ✓ The filter is simply a capacitor connected from the rectifier output to ground.
- ✓ RL represents the equivalent resistance of a load.



(a) Initial charging of the capacitor (diode is forward-biased) happens only once when power is turned on.



(b) The capacitor discharges through R_L after peak of positive alternation when the diode is reverse-biased. This discharging occurs during the portion of the input voltage indicated by the solid blue curve.



(c) The capacitor charges back to peak of input when the diode becomes forward-biased. This charging occurs during the portion of the input voltage indicated by the solid blue curve.

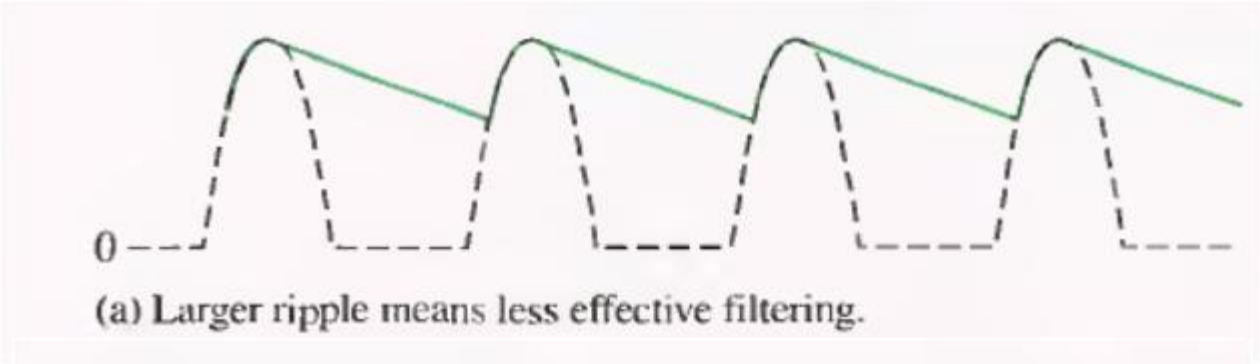
- ✓ During the positive first quarter-cycle of the input, the diode is forward-biased, allowing the capacitor to be charged as shown in fig (a).

$$v_c = v_m\text{-}0.7\text{v}$$

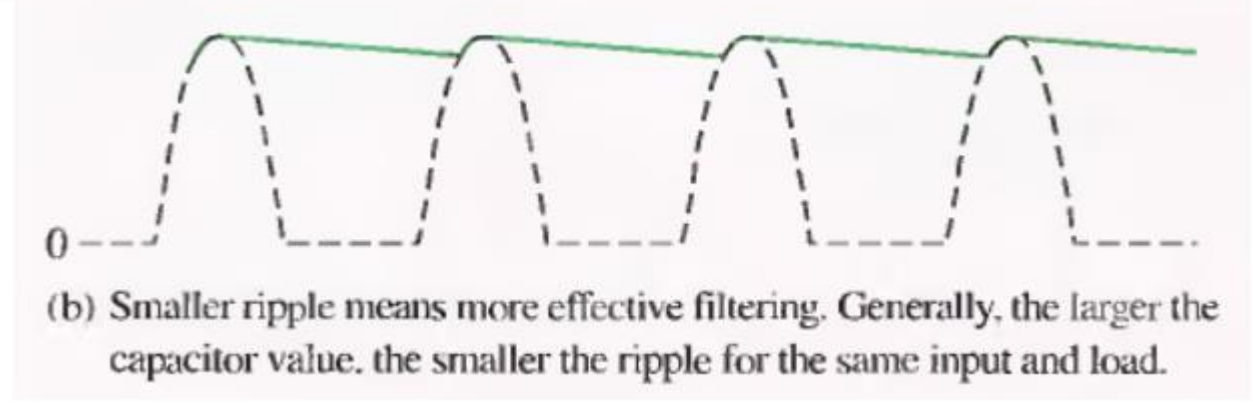
- ✓ When the input begins to decrease below its peak. as shown in part (b), the capacitor retains its charge and the diode becomes reverse-biased because the cathode is more positive than the anode.
- ✓ During the remaining part of the cycle, the capacitor can discharge only through the load resistance at a rate determined by the RLC time constant, which is normally long compared to the period of the input.
- ✓ The larger the time constant, the less the capacitor will discharge.
- ✓ During the first quarter of the next cycle, as illustrated in part (c), the diode will again become forward-biased when the input voltage exceeds the capacitor voltage by approximately 0.7 V.

Ripple Voltage

- ✓ Capacitor quickly charges at the beginning of a cycle and slowly discharges through RL after the positive peak of the input voltage (when the diode is reverse-biased).
- ✓ The variation in the capacitor voltage due to the charging and discharging is called the ripple voltage.
- ✓ Generally, ripple is undesirable; thus, the smaller the ripple, the better the filtering action.



(a) Larger ripple means less effective filtering.



(b) Smaller ripple means more effective filtering. Generally, the larger the capacitor value, the smaller the ripple for the same input and load.

Half wave and full wave ripple voltage

- ✓ The frequency of a full-wave rectified voltage is twice that of a half-wave rectified voltage.

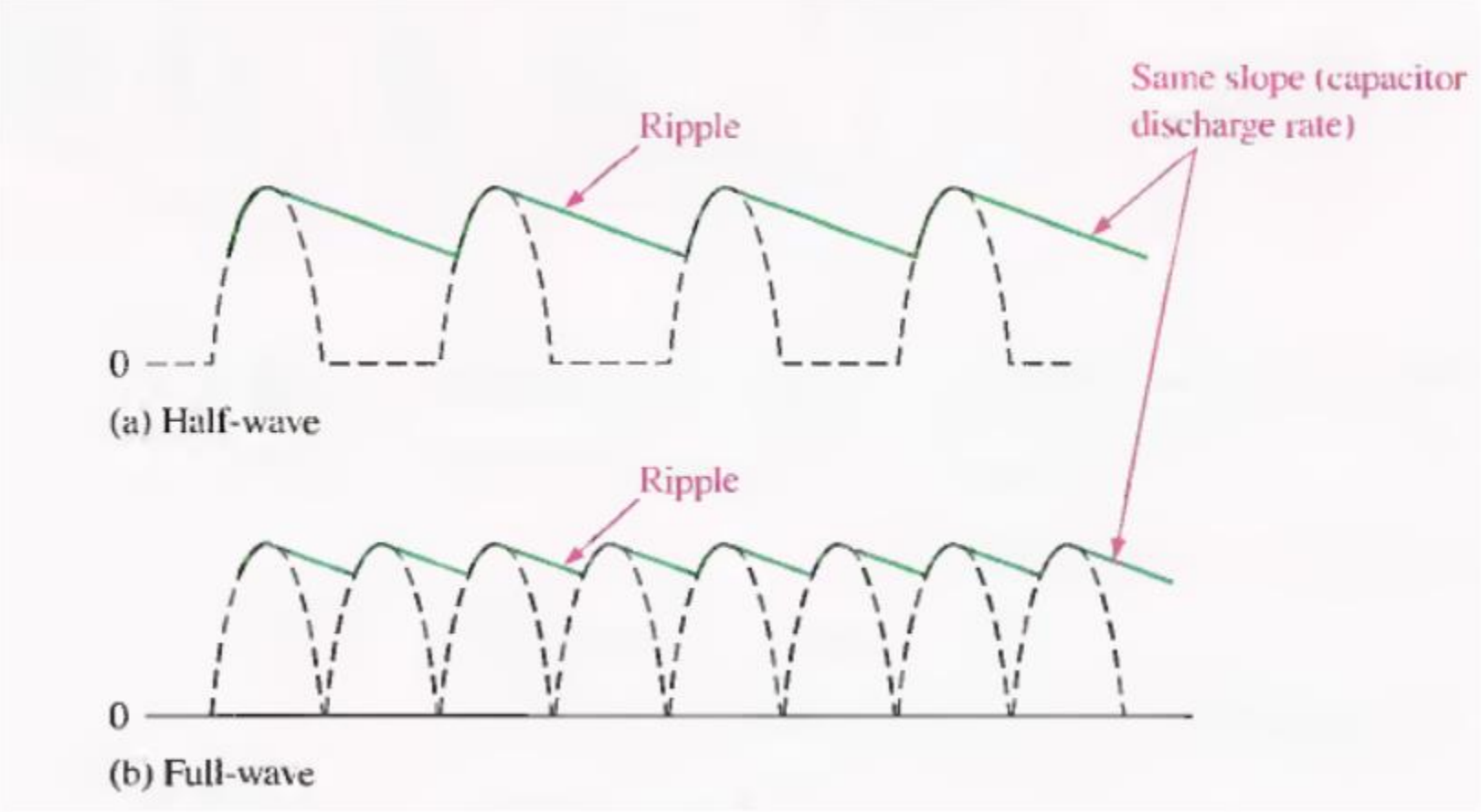
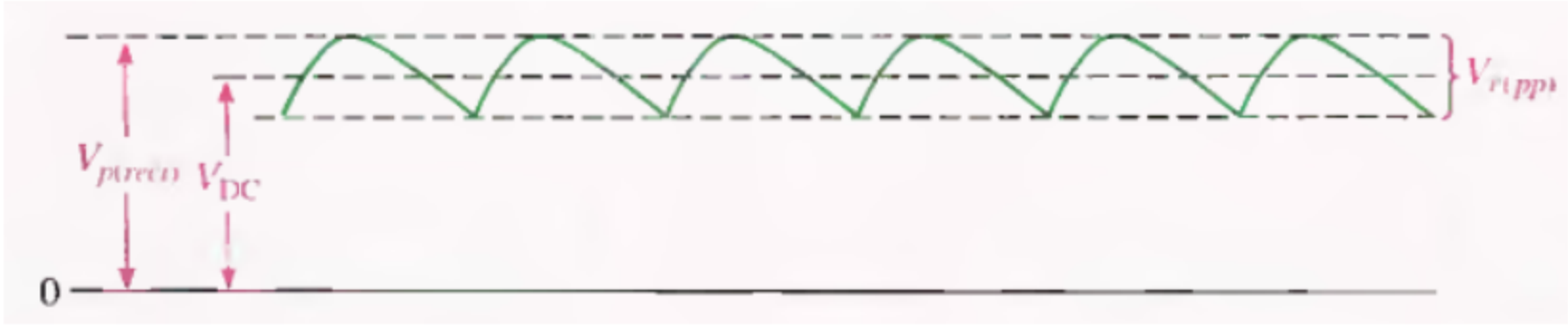


Fig. Comparison of ripple voltages for half-wave and full-wave rectified voltages with the same filter capacitor and load and derived from the same sinusoidal input voltage.

Ripple Factor



✓ The ripple factor (r) is an indication of the effectiveness of the filter and is defined as

$$r = \frac{rms\ ripple\ voltage}{dc\ (average\ voltage)} \times 100\%$$

where $v_{rms} = \frac{v_m}{\sqrt{2}}$

$v_{dc} = 0.636 v_m$ after rectification

Voltage Regulators

- ✓ The combination of a large capacitor and a voltage regulator helps produce an excellent power supply.
- ✓ Most regulators are integrated circuits and have three terminals:- an input terminal, an output terminal and a reference (or adjust) terminal.
- ✓ The input to the regulator is first filtered with a capacitor to reduce the ripple to < 10%.
- ✓ The regulator reduces the ripple to a negligible amount.
- ✓ voltage regulators can furnish a constant output of one or more amps of current with high ripple rejection.

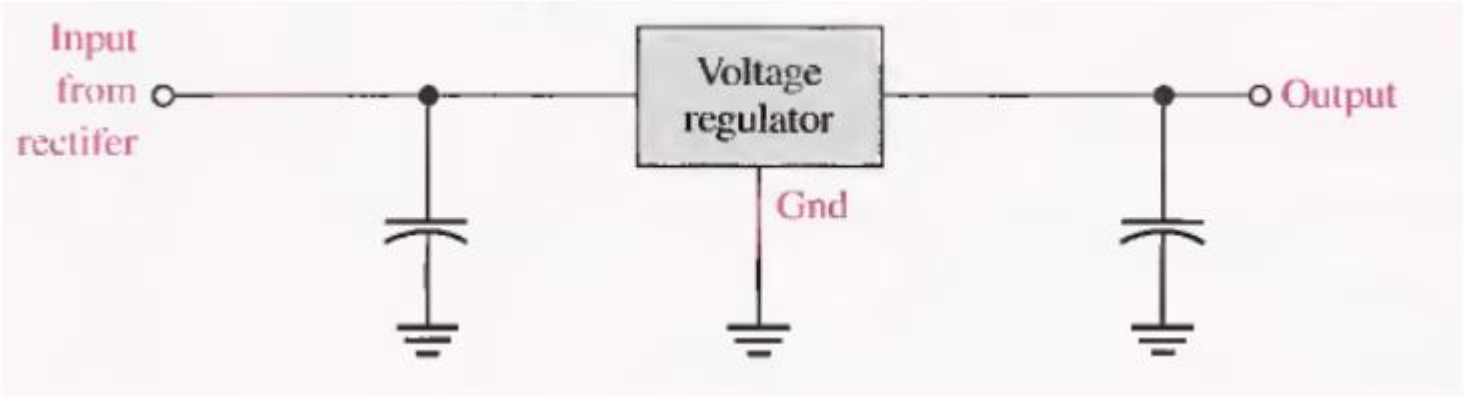


Fig. A voltage regulator with input and output capacitors.

Percent Regulation

Line regulation

- ✓ specifies how much change occurs in the output voltage for a given change in the input voltage.
- ✓ It is typically defined as a ratio of a change in output voltage for a corresponding change in the input voltage expressed as a percentage.

$$\text{Line regulation} = \left(\frac{\Delta v_{out}}{\Delta v_{in}}\right) * 100\%$$

Load regulation

- ✓ specifies how much change occurs in the output voltage over a certain range of load current values, usually from minimum current (no load, NL) to maximum current (full load, FL).

✓ It is normally expressed as a percentage and can be calculated with the following formula:

$$\text{Load regulation} = \left(\frac{v_{NL}-v_{FL}}{v_{FL}}\right)* 100\%$$

where

v_{NL} is the output voltage with no load and v_{FL} is the output voltage with full (maximum) load.

EX-

A certain regulator has a measured no-load output voltage of 5.18 V and a full- load output of 5.15 V What is the load regulation expressed as a percentage?

Solution

$$\text{Load regulation} = \left(\frac{v_{NL}-v_{FL}}{v_{FL}}\right)* 100\% = \left(\frac{5.18V-5.15V}{5.15V}\right)*100\% = 0.58\%$$

Related Problem

If the no-load output voltage of a regulator is 24.8 V and the full-load output is 23.9 V. what is the load regulation expressed as a percentage?

Special purpose diodes

Zener diode

- ✓ A zener diode is a silicon pn junction device that is designed for operation in the reverse-breakdown region.

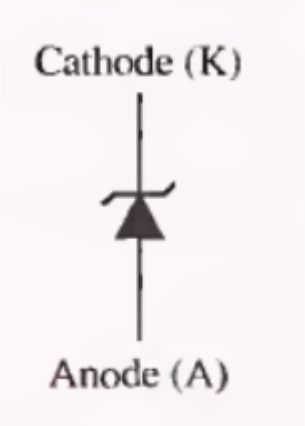


Fig. Zener diode symbol.

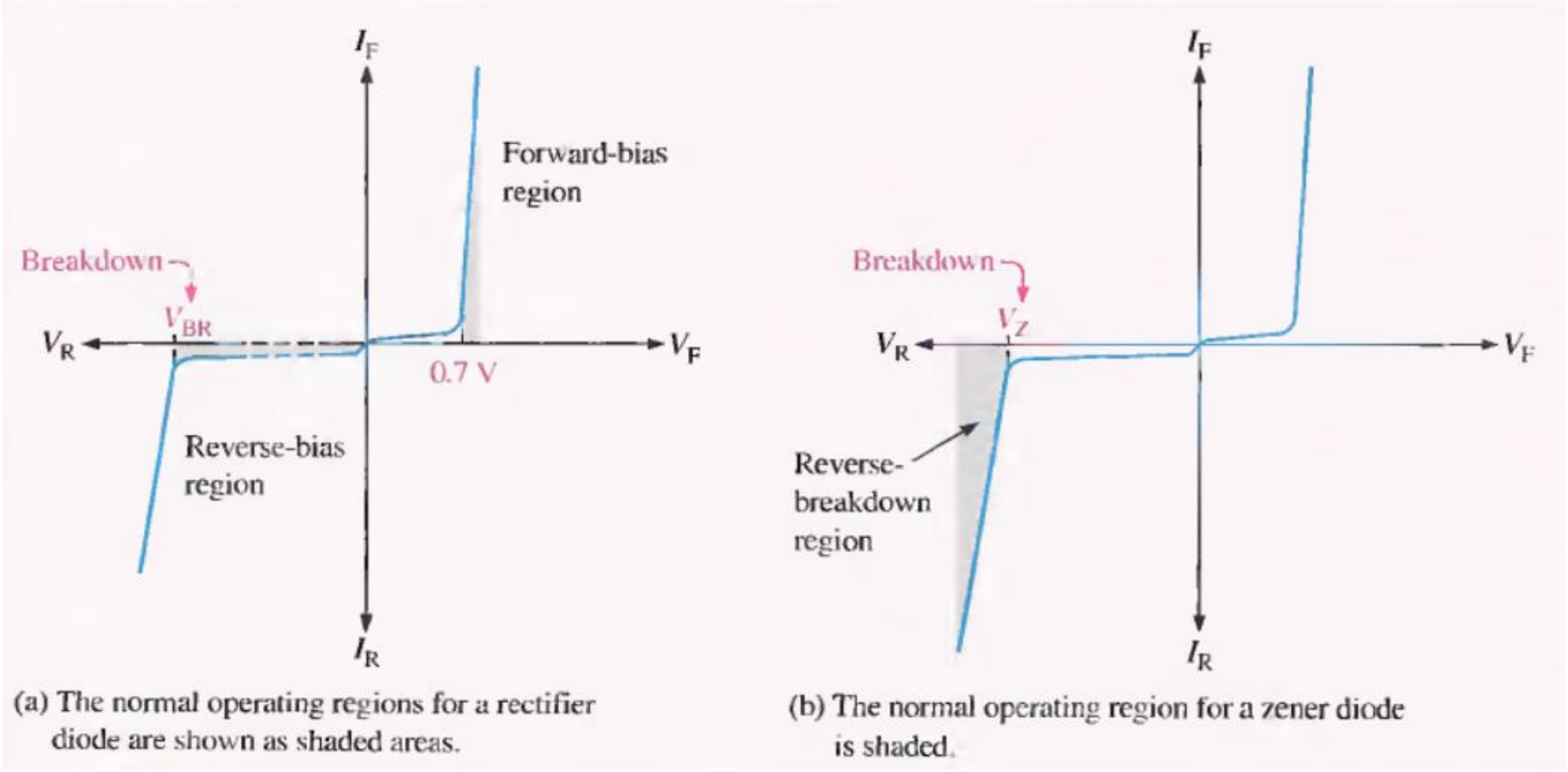


Fig. General diode V-I characteristic.

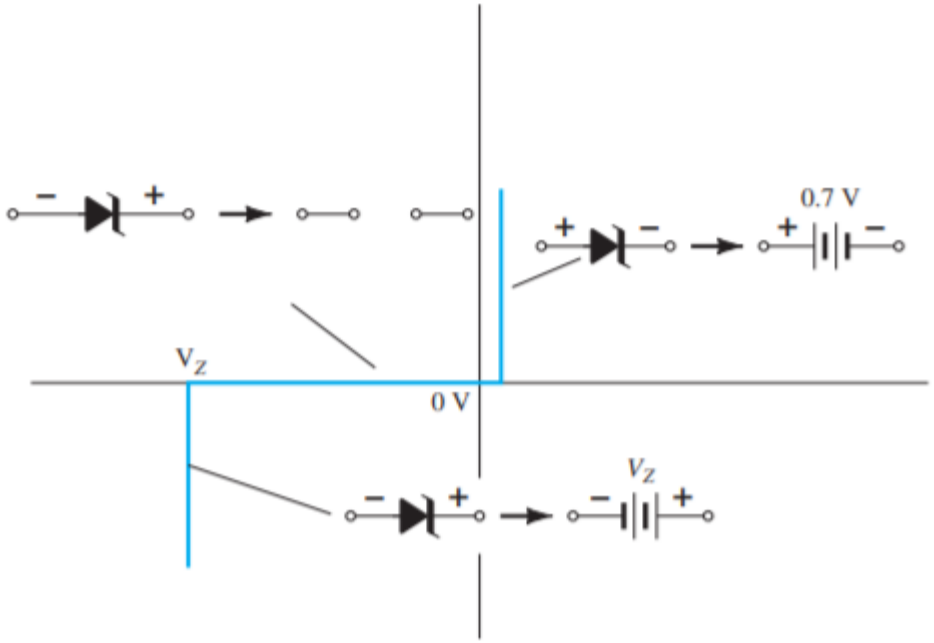


Fig. Approximate equivalent circuits for the Zener diode in the three possible regions of application.

Optical diodes

Light emitting diode (LED)



Symbol for an LED. When forward- biased, it emits light.

- ✓ light-emitting diode is a diode that gives off visible or invisible (infrared) light when energized.
- ✓ When the device is forward biased, electrons cross the pn-junction from the n-type material and recombine with holes in the p-type material.
- ✓ This recombination requires that the energy possessed by the unbound free electrons be transferred to another state (CB).
- ✓ When recombination takes place, the recombining electron releases energy in the form of heat and light.

- ✓ Greater percentage of the energy dissipated (converted) during recombination for Si and Ge diodes is in the form of heat within the structure, and the emitted light is insignificant.
- ✓ For this reason, silicon and germanium are not used in the construction of LED devices.
- ✓ On the other hand, Diodes constructed of GaAs emit light in the infrared (invisible) zone during the recombination process at the p–n junction.
- ✓ But, a diode constructed of GaAsP emits visible light.
- ✓ Even though the light is not visible, numerous applications of infrared LEDs includes:
    - security systems
    - industrial processing
    - optical coupling
    - safety controls such as on garage door openers, and in home entertainment centers, where the infrared light of the remote control is the controlling element.

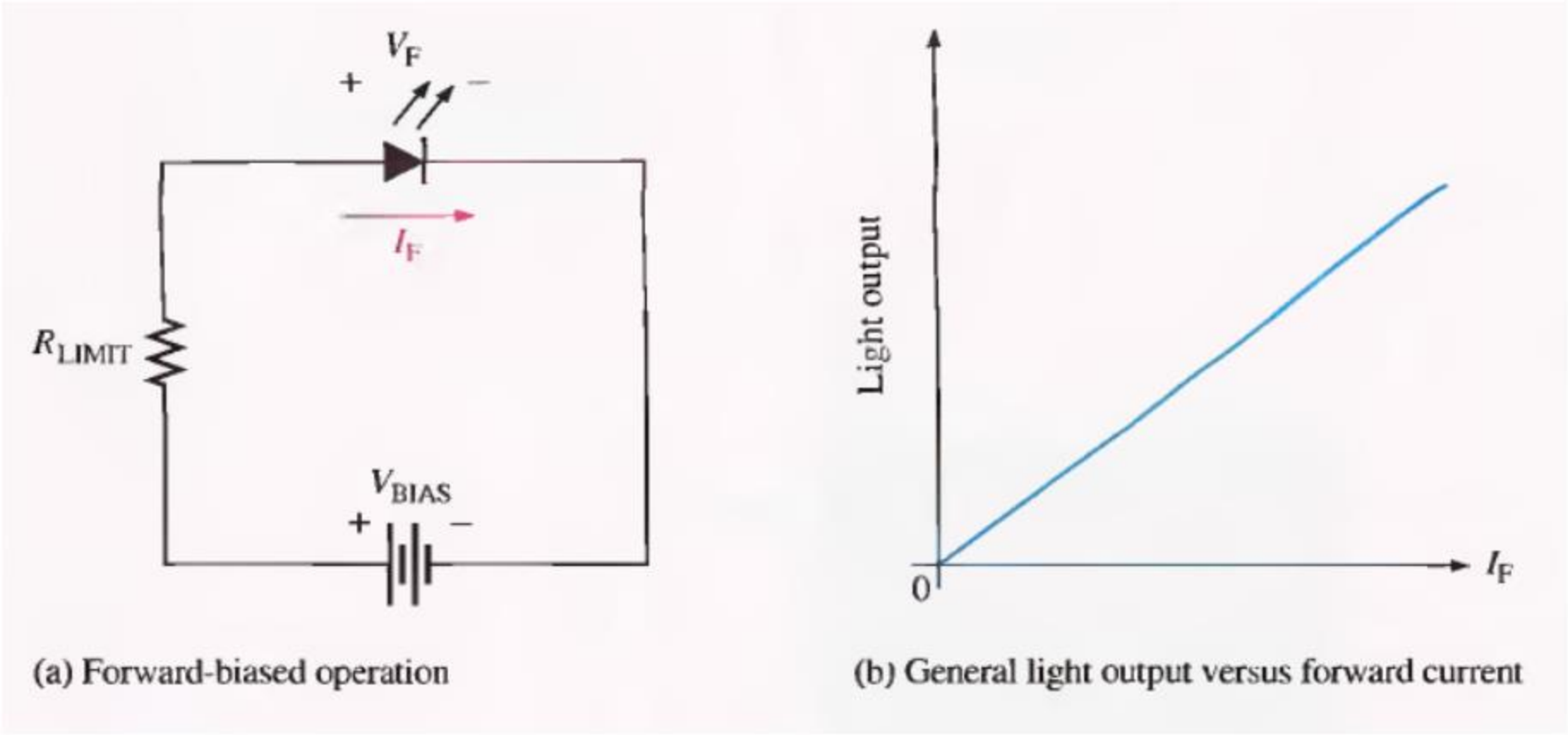
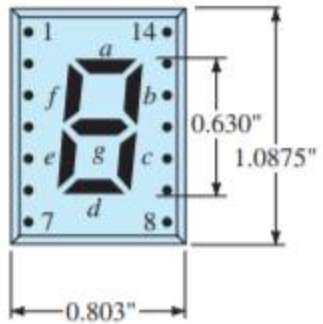
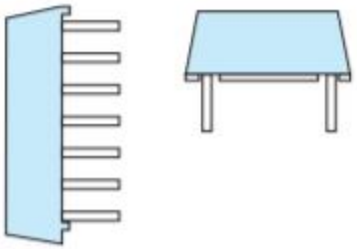


Fig. Basic operation of an LED.



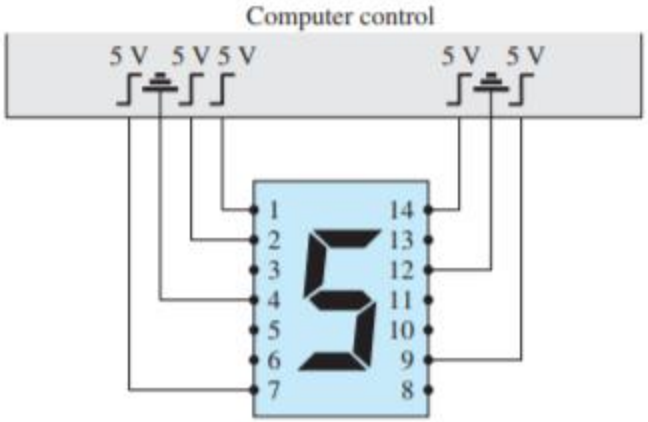
(a)



COMMON CATHODE

PIN #	FUNCTION
1.	Anode f
2.	ANODE g
3.	NO PIN
4.	COMMON CATHODE
5.	NO PIN
6.	ANODE e
7.	ANODE d
8.	ANODE c
9.	ANODE d
10.	NO PIN
11.	NO PIN
12.	COMMON CATHODE
13.	ANODE b
14.	ANODE a

(b)



(c)

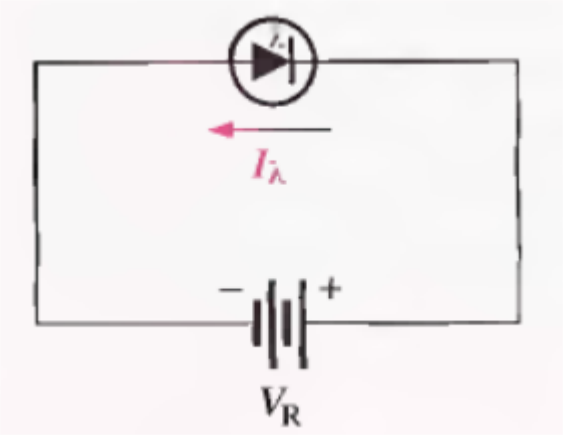
Fig. Seven-segment display: (a) face with pin idenfication; (b) pin function; (c) displaying the numeral 5

The Photodiode



Fig. Photodiode symbol

✓ Photodiode is a device that operates in reverse bias.



$I_\lambda - reverse\ current$

Fig. Reverse-bias operation

- ✓ The photodiode has a small transparent window that allows light to strike the pn-junction.
- ✓ The reverse-biased current is produced by thermally generated electron-hole pairs in the depletion region. which are swept across the pn junction by the electric field created by the reverse voltage.
- ✓ In a rectifier diode, the reverse leakage current increases with temperature due to an increase in the number of electron-hole pairs.
- ✓ Additional minority carriers are generated by light energy, so the greater the light intensity, the greater the reverse current, and the smaller the effective resistance.
- ✓ A photodiode differs from a rectifier diode in that when its pn junction is exposed to light, the reverse current increases with the light intensity.
- ✓ When there is no incident light, the reverse current, I_λ is almost negligible and is called the dark current.

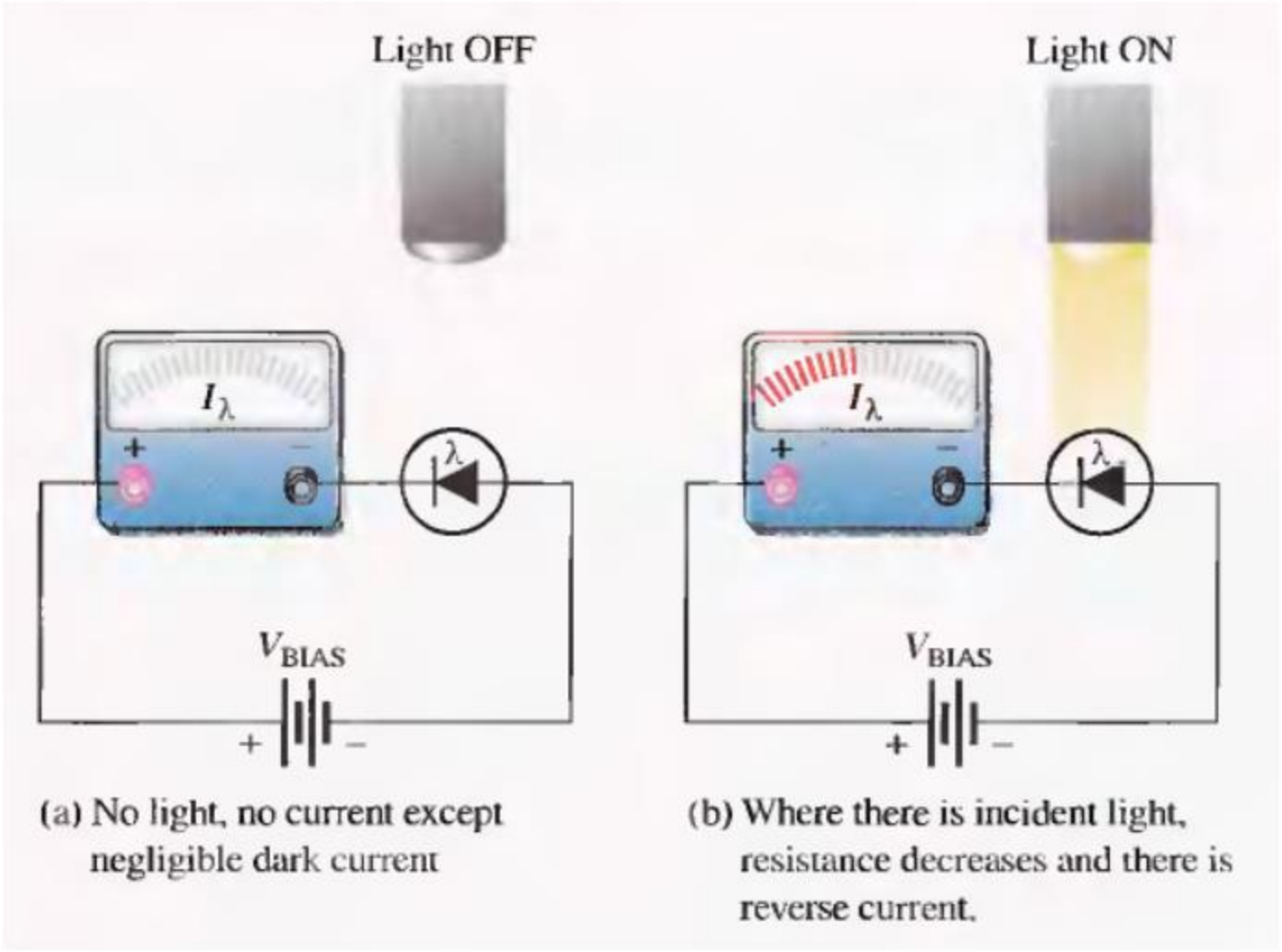


Fig. Operation of a photodiode.

The Schottky Diode



Fig. Schottky diode symbol.

- ✓ Schottky diodes are used primarily in high-frequency and fast-switching applications.
- ✓ A Schottky diode is formed by joining a doped semiconductor region (usually n-type) with a metal such as gold, silver, or platinum.
- ✓ Rather than a pn junction, there is a metal-to-semiconductor junction, as shown in Figure below.

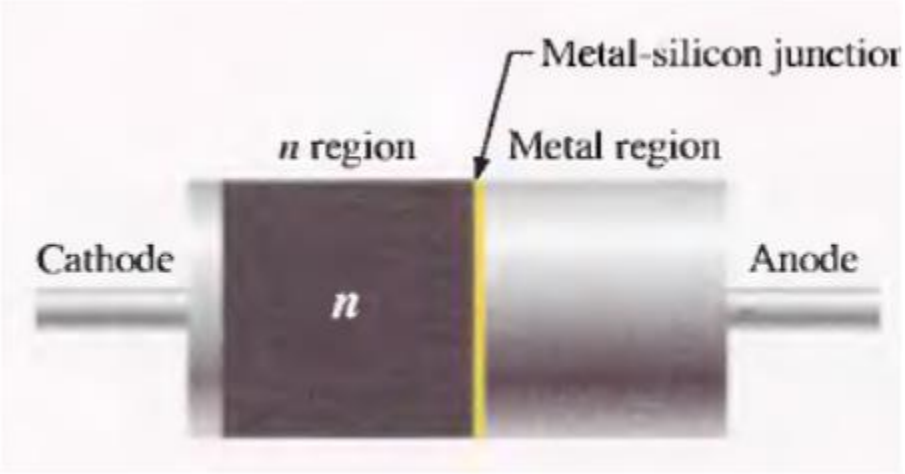


Fig.
Basic internal construction of a Schottky diode.

- ✓ The Schottky diode operates only with majority carriers.
- ✓ There are no minority carriers and thus no reverse leakage current as in other types of diodes.
- ✓ The metal region is heavily occupied with conduction-band electrons, and the n-type semiconductor region is lightly doped.
- ✓ When forward-biased, the higher energy electrons in the n region are injected into the metal region where they give up their excess energy very rapidly.
- ✓ Since there are no minority carriers, as in a conventional rectifier diode, there is a very rapid response to a change in bias.
- ✓ The Schottky is a fast-switching diode, and most of its applications make use of this property.
- ✓ It can be used in high-frequency applications and in many digital circuits to decrease switching times.